

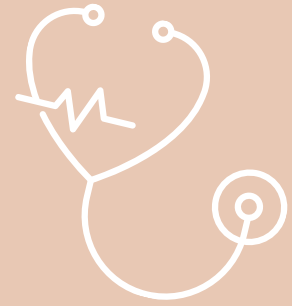


Nclex study guide pdf

General Biology (Evergreen Valley College)



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NCLEX STUDY GUIDE

Everything you need to know to pass the
NCLEX

ETSY - NURSINGSTUDENTGUIDES

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Advice for NCLEX

Plan to study for the NCLEX for at LEAST one month

The KEY for passing NCLEX is taking as many test questions as you can

Knowledge of disease processes and pharmacology is important, but it is more important to understand what the question is really asking.

Become very comfortable with Select All That Apply (SATA) 7

- 75% of my exam was SATA

Test banks and resources I recommend:

- U-World
 - #1 choice if you had to pick anything
 - Read the entire rationale for each question, even if you got it right
- Nurse Achieve
 - CAT (Computerized Adaptive Testing) format which is the exact testing format used on the NCLEX
 - Highly recommend taking a few CAT exams one to two weeks before your scheduled NCLEX date
- Saunders Comprehensive Review for the NCLEX-RN
 - Testing bank
 - Book provides a comprehensive review of all testing
 - Materials on NCLEX
 - I only recommend purchasing the book if you graduated from your nursing program a while ago and do not remember a lot of material

NCLEX STUDY SCHEDULE

Month:

Mon	Tues	Wed	Thru	Fri	Sat	Sun
Med Surg						
Critical Care						Review previous concepts Test Questions
Pediatrics						Review Test Questions
Maternity						Review Test Questions

NCLEX STUDY SCHEDULE

Month:

Mon	Tues	Wed	Thru	Fri	Sat	Sun
Mental Health				Leadership		Review Test Questions
Pharmacology			Test Questions Review topics that you don't feel confident in			
			NCLEX			

Must Know

Labs

K	3.5 - 5.0 mEq/L
Na	135 - 145 mEq/L
Ca	8.5 - 10.5 mEq/L
Cl	95 - 105 mEq/L
Mg	1.5 - 2.5 mg/dL
Phos	2.5 - 4.5 mg/dL
Fe	50 - 175 ug/dL
RBC	4.5 - 5.0 mEq/L
WBC	4.5K - 10K cells/mcL
Plt	150k - 400k cells/mcL
Hgb	13.5 - 16.5 (male) g/dL 12 - 15 (female) g/dL
Hct	41 - 50% (male) 35 - 44% (female)
BUN	7 - 20 mg/dL
Cr	0.7 - 1.4 mg/dL
Specific Gravity	1.010 - 1.030
Albumin	3.5 - 6.0 g/dL
Protein	6.2 - 8.1 g/dL
Bilirubin	<1.0 mg/dL
LDH	<100 mg/dL
HDH	>60 optimal mg/dL

Triglycerides	<150 mg/dL
Total cholesterol	130 - 200 mg/dL
INR	0.8 - 1.2 2 - 3.5 if on warfarin
PT	10 - 13 s
PTT	25 - 35 s Heparin: 1.5 - 2.5x normal
Troponin	< 0.035 ng/mL
GFR	>90 mL/min/1.73 m ²
Glucose	70 - 115 mg/dL
HBA1C	<7%
pH	7.35 - 7.45
CO2	35 - 45 mEq/L
HCO3	22 - 26 mEq/L
PaO2	80 - 100%

Vital Signs

BP	120/80
HR	60 - 100 bpm
SpO2	95 - 100%
RR	12 - 20 rpm
Temp	36.5 - 37.5° C

INR

Normal range: < 1.1	< 2 = risk for developing clots
Warfarin range: 2-3	>3 = risk for excessive bleeding d/t prolonged clotting time

Key

NI- nursing interventions
S/sx - signs and symptoms
Pt - patient
UO - urine output

Must Know

PT - Prothrombin Time

- Blood test that assesses how FAST blood clots
- Dependent on vitamin K (made by the liver) to make clots
- High levels caused by: vitamin K deficiency, cancer, liver disease

INR

- Lab used for patients who are taking warfarin
- INR level determines amount of warfarin given

Warfarin

- Measures the speed of clotting using extrinsic pathway

PTT - Partial Thromboplastin Time

- Lab used to assess if blood-thinning therapy (heparin) is working

Intracellular - fluid inside of a cell

Extracellular - fluid outside of cells

- Includes: interstitial fluid, connective tissue, bone, blood

Intravascular - fluid inside a blood vessel

Isotonic

- No osmotic pressure; cells stay same size
- 0.9% NS, Lactated Ringers, D5W

Hypotonic

- Water enters the cells. *Assess for edema*
- 0.45% NS, 0.33% NS

Hypertonic

- Water is removed from cells
- 3% and 5% NS, D5W with $\frac{1}{2}$ NS, D10W, D5LR

Colloid

- Fluid moves from interstitial to intravascular compartment. *Given to patients in severe hypovolemia*
- Albumin, Dextran

Beta 1 and Beta 2

- Beta 1 - found in heart
- Beta 2 - found in lungs
 - Remember: 1 heart, 2 lungs

Electrolyte Level Relationships

Inverse	Similar
Sodium & Potassium - low Na = high K	Potassium & Magnesium - low K = low Mg
Phosphorus & Magnesium - low Phos = high Mg	Calcium & Vitamin D - low calcium = low Vit D
Calcium & Phosphorus - low Ca = high Phos	Magnesium & Calcium - low Mg = low Ca

Must Know

Acid Base Balance

	Acidosis	Alkalosis
pH	7.35	7.45
PaCO ₂	45	35
HCO ₃	22	26

From the butt (diarrhea) - Metabolic Acidosis

From the mouth (vomitus) - Metabolic Alkalosis

ALKALOSIS: K is LOW

Acidosis: K is HIGH

- Hyperventilation increases chance of respiratory alkalosis due to increase of carbon dioxide in the blood
- Alkalosis: too much bicarb or loss of acid from the blood

Antidotes

Heparin → Protamine Sulfate

Warfarin (Coumadin) → Vitamin K

Potassium → Insulin, Kayexalate

Magnesium sulfate → Calcium Gluconate

Tylenol (Acetaminophen) → Mucomyst (acetylcysteine)

Opiates → Narcan

Narcotics → Narcan

Digoxin → Digiband

Insulin → Glucose

Cholinergic crisis → Atropine and oximes

Iron → Deferoxamine

Beta Blockers → Glucagon

TPA → Aminocaproic acid

Benzodiazepines → Romazicon

Important Info

- NPO anytime there is a GI issue
- Pt needs 2-3L of fluid/day
- Give meds 1 hour before meal or 2 hours after meal
- Give antacids 1 hour before meds or 4 hours after meds
- Neuro pt → keep head of bed 30-45 degrees
- After surgery monitor: first 24 hours → bleeding. 48 hours → infection
- If pt is admitted for a fluid problem → check daily weights
- In patients with severe vomiting and diarrhea, monitor for electrolyte imbalances
- Anaphylaxis → give ordered epinephrine
- ALWAYS obtain cultures before starting antibiotics
- If pt has dysphagia → risk for aspiration pneumonia

When in doubt

- Electrolytes: choose potassium (K⁺)
- Choose an answer that has you stay with your patient
- Remove any answer choices that are absolutes (all/none)
- Pick the least invasive option first (if it is not an emergency)
- Emergency → choose STAT words. Choose the first thing you would do in order to save or avoid causing harm to the pt
- Never withhold treatment from a patient
- Empower your patient

Tell Tale Signs

- Older patient with acute onset confusion → UTI
- Restlessness, agitation, decreased level of consciousness → EARLY SIGN of deterioration

Hallmark Signs

Addison's - bronze skin

Anaphylaxis- tachycardia & wheezing

Angina - sharp chest pain

Appendicitis - rebound tenderness

Asthma - expiratory wheezing

Bulimia - chipmunk cheeks

BPH - polyuria at night

Cataract - blurred vision

Cushing's - moonface/buffalo hump

Cystic fibrosis - salty skin

Cystitis - burning urination

Diabetes Mellitus - polyuria, polyphagia, polydipsia

DKA - Kussmaul's breathing

DVT - Homan's sign

Emphysema - barrel chest

Epiglottitis-Dysphagia,dysphonia, drooling

Glaucoma - halo vision; painful

Guillain Barre - ascending weakness

Hyperglycemia - acetone breath

Hyperthyroidism - bulging eyes

Hypocalcemia - Chvostek and Trousseau sign

IBS - GI pain, diarrhea, constipation

ICP (elevated) - Cushing's Triad (HTN, bradycardia, bradypnea)

Intussusception - sausage-shaped mass on R side

Kawasaki Syndrome - strawberry tongue

Leprosy - lioning face

Liver cirrhosis - spider veins

Lupus - butterfly rash

Lyme Disease - Bull's eye rash

Measles - Koplick's spots

Meningitis - Brudzinski and Kernig's sign

Myasthenia Gravis - descending weakness

Myocardial infarction - chest pain, radiated to left arm and neck

Pancreatitis - Cullen's sign

Parkinson's - tremor

Pernicious anemia - beefy red tongue

Pneumonia - rust colored sputum

Sepsis- tachycardia, hypotension, SOB

Shock - tachycardia, tachypnea, hypotension

TEF - cyanosis, choking, coughing, continuous drooling

Ulcerative colitis - frequent bloody diarrhea

Patient Positioning

- Epidural puncture → side lying
- During lumbar puncture → lateral recumbent fetal position
- After lumbar puncture → supine (to prevent headache and leaking of CSF)
- Pulmonary embolism → turn to left side and lower HOB
- Shortness of breath → high fowlers
- Lung biopsy → lay on side of the bed or with arms raised up on pillows over bedside table
- Chest tube insertion → arm raised above head
- NG tube placement → high fowlers
- Tube feeding with decreased LOC → put on right side (promotes emptying of stomach) with HOB elevated (prevent aspiration)
- Prevent dumping syndrome → eat in low-fowlers, lie down after meals for 30 min
- Hiatal hernia → Fowler or semi-Fowler (reduce regurgitation)
- After thyroidectomy → low semi-fowlers
- After myringotomy → lay on affected ear (allows drainage)
- After cataract surgery → lay on unaffected side
- During continuous bladder irrigation → keep leg straight
- During internal radiation → on bedrest while implant in place
- Pt with heat stroke → lie flat with legs elevated
- After total hip replacement → sleep on unaffected side, don't elevate HOB more than 45 degrees, maintain hip abduction by separating thighs with pillow
- Buck's Traction → elevate foot of bed for counter traction
- Above and below knee amputation → elevate for first 24 hours, position prone daily for hip extension
- Back pain → William's position
- Myocardial infarction → semi-fowlers
- Autonomic dysreflexia/hyperreflexia → elevate HOB
- Head injury → elevate HOB 30 degrees (decrease intracranial pressure)

Patient Positioning

- After supratentorial surgery (incision behind hairline) → elevate HOB 30-45 degrees
- After infratentorial surgery (incision at nape of neck) → lay flat and lateral on either side
- Peritoneal dialysis when outflow is inadequate → turn pt side to side then check for kinks in tubing
- Administering an enema → lay on left side with knees flexed (Sim's)
- After appendectomy → right side with legs flexed
- During paracentesis → High fowler's (HOB 45-90 degrees)
- During thoracentesis → sit at edge of bed, lean forward, and rest arms on table
- Spina bifida → prone (so sac doesn't rupture)
- Infant with cleft lip → side, supine, or upright in infant seat to prevent ripping of suture line.
- Infant with ONLY cleft palate → side or abdomen
- Woman in labor with unreassuring FHR → turn on left side
- Prolapsed cord → knee to chest or Trendelenburg

Diagnostic Signs/Tests

Allen's Test

- Assesses arterial blood supply in hand. Positive test means the pt does NOT have adequate blood supply to the hand
- How to do: block both radial and ulnar arteries until skin becomes pale, then release ulnar. If hand becomes pink, ulnar arter works, and a ABG/radial stick can be done

Cullen's Sign

- Internal hemorrhage causing ecchymosis around umbilical region; seen in pancreatitis

Turner's sign

- Internal hemorrhage seen as grayish blue on your flanks; pancreatitis

Murphy's Sign

- Pain with palpation of gallbladder area; cholecystitis

McBurney's Point- RLQ pain indicative of appendicitis

LLQ Pain- diverticulitis; make sure pt has no seeds, nuts, peas

RLQ Pain- appendicitis; assess for peritonitis

Guthrie Test

- Neonatal heel prick that tests for PKU (phenylketonuria)

Babinski Sign- assessment for nervous system issues done by stimulating bottom of foot

- Negative: toes curl
 - Normal in adults and children older than 2 years, abnormal in children 2 years and under
- Positive: toes fan
 - Abnormal in adults and children older than 2 years, normal in children 2 years and under

Beck's Triad

- Indicative of cardiac tamponade - hypotension, JVD, muffled heart sounds

Halo Sign

- Positive - appears as concentric rings (bloody circle surrounded by yellow border)
- Fluid from head injury contains cerebrospinal fluid (CSF); fluid also contains glucose

Endoscopic Retrograde Cholangio Pancreatography (ERCP)

- Procedure to examine diseases of the pancreas, liver, gallbladder, and bile ducts
- Preop - NPO. Postop- assess gag reflex and s/sx of perforation

Additional Notes

- If at home: Call 911 if chest pain is not relieved by nitroglycerine; keep med in original container and away from light
- Definitive diagnosis for abdominal aortic aneurysm → CT scan

General Nursing

#1 Priority- ABC's

Airway

Breathing

Circulation

Maslow's Hierarchy

- ABC's
- Safety
- Comfort (Pain)
- Psychological
- Social
- Spiritual

When in distress, DO NOT ASSESS!

Fire Safety

- RACE - rescue patients at risk, activate alarm, contain fire by closing doors and windows, extinguish fire if able

Prevention Types

Primary

- Preventing disease or injury
- Ex: educating on first aid

Secondary

- Treating patients who are injured
- Ex: emergency department, triage

Tertiary

- Follow-up care

Nursing Process:

1. Assessing - gather information
2. Diagnosing - name the problem
3. Outcome/Planning - develop plan of care
4. Implementing - perform nursing interventions
5. Evaluating - did interventions work?

Order of Assessment

Adults	Abdomen/Children
Inspection	Inspection
Auscultation	Auscultation
Palpation	Percussion
Percussion	Palpation

*In kids, go from least to most invasive

Pulse Strength

- 0 absent, 1+ weak, 2+ normal, 3+ bounding

Arterial Blood Gases

- Before drawing an ABG, perform the Allen's Test to check for sufficient blood flow
- ABG needs to be in a heparinized tube (blocks clotting cascade)
- Put on ice immediately after drawing

General Nursing

Burns

- #1 priority: Maintain a patent airway
- Rule of 9s
 - 9= head
 - 18= arms
 - 36= torso
 - 36= legs
 - 1= perineum
- Parkland Formula
 - Calculation for the total fluid requirement 24 hours after a burn
 - $4\text{ml} \times \text{Total Burn Surface Area} \times \text{body weight (kg)} = \text{Total fluid pt will receive in 24 hrs}$
 - 50% given in the first 8 hours
 - 50% given in the next 16 hours
- 1st Degree- red and painful
- 2nd Degree- blisters and edema
- 3rd Degree- no pain due to nerve damage

Burn Types

1st Degree Superficial thickness	2nd Degree Superficial partial thickness	2nd Degree Deep partial thickness	3rd Degree Full thickness	4th Degree Deep full thickness
Damages epidermis Pink/red, no blisters Heals 3-7 days	Damages dermis Pink/red, blisters, edema Heals <21 days	Damages dermis Red/white, edema, wound balances w/ pressure Heals >21 days	Total destruction of epidermis + dermis Skin is deep red, brown, black, or white Skin graft needed for healing	Damage extends to muscle, bone, tendon Skin is black Skin graft or amputation

NI: airway patency (intubation may be necessary), give O2, assess vital signs, give IV fluids, assess for paralytic ileus, pain management, assess for s/sx of infection

Glasgow Coma Scale

BEHAVIOR	RESPONSE	SCORE
Eye Opening Response	Spontaneously	4
	To speech	3
	To pain	2
	No response	1
Best Verbal Response	Oriented to time, place, and person	5
	Confused	4
	Inappropriate words	3
	Incomprehensible sounds	2
Best Motor Response	No response	1
	Obey commands	6
	Moves to localized pain	5
	Flexion withdrawal from pain	4
Total Score	Abnormal flexion	3
	Abnormal extension	2
	Best response	15
	Comatose client	8 or less
	Totally unresponsive	3

General Nursing

Cranial Nerves

1. Oh (olfactory I)	smell
2. Oh (optic II)	vision
3. Oh (oculomotor III)	constricts pupils
4. To (trochlear IV)	moves eyes up and down
5. Touch (trigeminal V)	corneal (blink) reflex
6. And (abducens VI)	abducts eyes (inward and outward)
7. Feel (facial VII)	smiling, blowing out cheeks
8. Very (vestibulocochlear/auditory VIII)	hearing and equilibrium
9. Good (glossopharyngeal IX)	swallowing
10. Velvet (vagus X)	gagging and swallowing
11. Such (spinal accessory XI)	head turning and shoulder shrugging
12. Heaven (hypoglossal XII)	tongue movement

Cultural

- Jewish: no meat and milk together
- Greek: puts protective charms on baby's neck to avoid eny from others

Blood

- O- universal donor
- AB + is the universal recipient
- Patients who are Rh negative should NOT receive Rh positive blood

Blood Transfusions

- Stay with pt for first 15 minutes
- Multiple blood transfusions = risk for hyperkalemia
- Transfusion reactions s/sx: fever, SOB, hypotension, fever, dizziness
- If reaction occurs: STOP transfusion, start IV line with NS, call MD and blood bank, monitor pt for s/sx

Compartment Syndrome

- Emergency situation caused by increased pressure and restricted blood flow to an extremity, which results in pain and paresthesia
- Requires immediate action, as damage is irreversible after 4-6 hours
- 5 P's: pain, paresthesia, paralysis, pallor, pulselessness
- NI: notify MD, fasciotomy to relieve pressure, loosen cast

Sucking stab wound

- Dress wound and tape it on 3 sides so air can escape
- Do NOT use occlusive dressing
- Next steps: get chest tube tray, labs, and start an IV

Chest Tube

- If it is accidentally removed, use occlusive dressing taped on 3 sides (allows air to escape and prevents tension pneumothorax)
- If it becomes disconnected, do NOT clamp
 - Place the end of the tube in a container of sterile saline (or water if saline not available)
- Fluctuations = good (fluid moves up with each inspiration and down with expiration) mean the chest tube is working
- No fluctuations → check for kinked tubing, occlusions, or ask patient to change position
- Continuous bubbling when the chest tube is connected to suction → air leak

Safety in the patient's room

- Ensure all the wheels are locked and bed is in lowest position
- Having all 4 side rails up is a form of restraint (even in patients who are considered a fall risk)

General Nursing

Pressure Ulcer Stages

Stage 1	Stage 2	Stage 3	Stage 4	Unstageable
Intact skin Non-blanchable erythema	Partial thickness Open Pink/red	Full thickness Subcutaneous tissue may show NO bone, tendon, or muscle	Full thickness WITH visible bone, tendon, or muscle	Localized purple or dark red Slough or eschar

Don and Doff Order

Don	Doff
Gown	Gloves
Mask	Goggles
Goggles	Gown
Gloves	Mask

Precautions

Contact

Gloves, gown

*water and soap for c. diff

Mrs. Wee

M - multidrug resistant organism

R - respiratory infection

S - skin infections (herpes simplex, impetigo, scabies)

W - wound infections

E - enteric infection (c. diff)

E - eye infection



Droplet

Gloves, gown, face mask, eye shield

Spiderman

S - Sepsis, streptococcal pharyngitis, scarlet fever

P - Pneumonia, pertussis, parovirus B19

I - Influenza

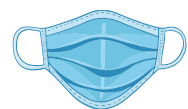
D - Diphtheria

E - Epiglottitis

R - Rubella

M - Meningitis, meningococcal pneumonia, mumps

An - Adenovirus



Airborne

Standard Precautions +
gloves, N95 mask

MTV

M - Measles

T - Tuberculosis

V - Varicella Zoster



Medical Surgical

Respiratory

Early sign of cerebral hypoxia → restlessness and irritability

Respiratory status

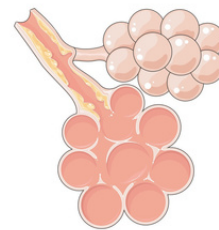
- Key is visualization and assessment of breath sounds
- If lungs sound clear and patient is blue, they are not receiving enough oxygen

Asthma

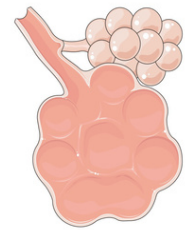
- Difficulty of breathing due to the narrowing, swelling, and production of mucus in the airway
- S/sx: SOB, wheezing, coughing, hypoxemia, respiratory acidosis
- NI: Administer bronchodilators and steroids, maintain hydration, deliver oxygen or nebulizer as prescribed
- Avoid morphine → histamine releasing opioid can lead to exacerbation
- Avoid NSAID's and aspirin - can worsen asthma symptoms

COPD - characterized by bronchospasm and dyspnea

- Bronchitis
 - Inflamed bronchioles, increased mucus
 - Blue bloaters, barrel chest
- Emphysema
 - Destruction of alveoli
 - Pink puffers
 - Encourage pursed-lip breathing (promotes CO₂ elimination), high-fowlers and leaning forward
- Normal SpO₂ for COPD: 88-92%. Do NOT raise SpO₂ level higher than 92% bc a low SpO₂ is what stimulates pt to breathe
- NI: offer mechanically soft foods (to save pt energy), teach pt to inject at least 3L of fluid/day (thins mucus)



Bronchitis



Emphysema

Empyema

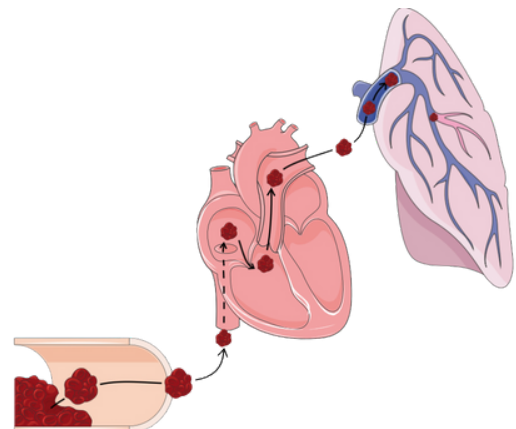
- Pus in the pleural cavity; associated w/ pneumonia or after thoracic surgery
- NI: elevate HOB, abx, chest tube or thoracentesis (drainage)

Acute Respiratory Distress Syndrome (ARDS)

- Fluid fills the alveoli in the lungs and inhibits oxygen exchange; causes severe hypoxemia
- First sign- increased respirations. Followed by dyspnea, retractions, cyanosis
- Hypoxemia is NOT responsive to O₂ therapy bc the fluid in the alveoli blocks the diffusion of oxygen
- NI: mechanical ventilation with PEEP

Pulmonary Embolism (PE)

- Thrombus gets lodged into pulmonary artery, blocking blood flow. Other cause is fat embolism from long bone fracture
- First sign - chest pain. Then dyspnea, tachypnea, blood tinged sputum
- Body compensates by hyperventilating → respiratory alkalosis
- NI: elevate HOB, give O₂, thrombolytic therapy, pain control, encourage ambulation (prevent venous stasis)



Respiratory

Pneumonia

- Infection resulting in decreased gas exchange in the affected lung lobes
- Alveoli become blocked with purulent fluid → impairs ventilation
 - Alveoli continue to receive perfusion from the pulmonary artery resulting in deoxygenated blood
 - Ventilation to perfusion (V/Q) mismatch or pulmonary shunt
 - May result in hypoxia and respiratory distress
 - Blood flow in the lungs is partially influenced by gravity, meaning blood flows in higher volumes to dependent parts of the lung
 - Unilateral pneumonia should be positioned with the unaffected (good) lung down to improve perfusion and oxygenation
 - *Ex: pt with a left lobar pneumonia: position in the right lateral position
 - Unaffected (good) lung down (right lung) to increase blood flow to the lung most capable of oxygenating blood
- Crackles heard on auscultation suggest pneumonia
- At risk: people 65 or older and infants under 2 years old (immune system still developing)
- Pneumococcal vaccine recommended for pts 65 years and older
- S/sx: pleuritic pain, wheezing, fever, sputum, change in LOC
- NI: Droplet precautions, O2 as needed, encourage deep breathing and coughing, increase fluid intake (thin mucus), abx, monitor LOC

V/Q Mismatch

Ventilation (airflow) or perfusion (blood flow) in the lungs is impaired

S/sx: SOB, fatigue, headache, confusion, dizziness, cyanosis

Pleural Effusion

- Fluid buildup between the lungs and chest; prevents lung expansion
- S/sx: pleuritic pain, dyspnea, dry cough, orthopnea
- NI: high Fowler's, monitor breath sounds, encourage deep breathing and coughing, prep for thoracentesis

Pleurisy

- Stabbing chest pain that usually increases on inspiration or with cough
- Caused by inflammation of the visceral pleura (over the lung) and the parietal pleura (over the chest cavity)
- The pleura space (between the 2 layers) normally contains about 10 mL of fluid to help the layers glide easily with respiration
- When inflamed, they rub together and cause pleuritic pain

Fremitus

- Palpable vibration felt on the chest wall
- Expected finding in pneumonia
- Sound travels faster in solids (consolidation) than in aerated lung → increased fremitus in pneumonia

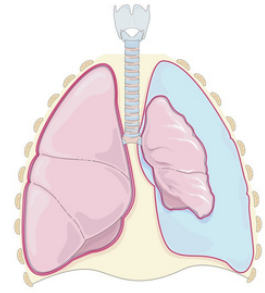
Tuberculosis

- PPD test: skin assessed 48-72 hours post administration
 - Positive if induration is: >15 mm in healthy pt or >10 mm in immunocompromised pts or >5 mm in high risk pts (HIV, recent contact with TB pt)
 - Positive and no symptoms → chest x-ray
 - Positive and symptomatic → sputum culture
 - Anyone who has received a bacillus Calmette-Guerin (BCG) vaccine will have a positive test and needs chest x-ray
- Airborne precautions, pt must wear mask if leaving room
- Public health risk; if pt does not comply with treatment, they need supervision
- Teaching is very important! Drug therapy is typically 6 months or longer
- Med: Rifapentine (Prifin) - may cause orange colored body secretions, take w/ meals

Respiratory

Pneumothorax

- Lung collapse due to air in pleural space (space b/w lungs and chest wall)
- Open pneumothorax - air circulates freely into pleural space
- Closed - air in pleural space does not increase
- Tension pneumothorax - air cannot leave pleural space; compresses lungs and shifts the mediastinum
- S/sx: sharp chest pain, SOB, cyanosis, tachycardia, tachypnea, hypotension
- NI: 3 way dressing, oxygen, chest tube, surgery (if needed)



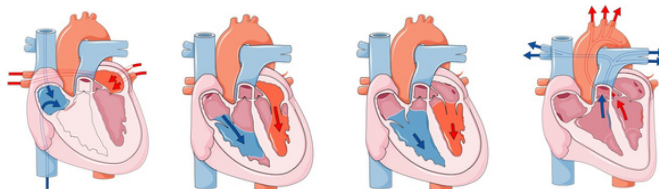
Nasopharyngeal airway

- Tube-like device used to maintain upper airway patency
- Used in alert or semiconscious patients; less likely to cause gagging
- NEVER insert them in a pt who may have had a head trauma (which might occur during a seizure) bc if they have a skull fracture, it may be malpositioned into structures and tissues in the brain
- CT must be done first to rule out fracture
- Size- measure tip of nose to the earlobe; select diameter smaller than naris

Cardiovascular

Blood Flow

- Tricuspid → Pulmonic → Mitral → Aortic
 - Toilet Paper My Ass



Listening to Heart Sounds

- APE To Man
- Aortic valve, pulmonary valve, Erb's point, tricuspid valve, mitral valve

Acute Coronary Syndromes

- Unstable Angina
 - Minor occlusion → pain
 - Typically relieved by meds
 - PCI (percutaneous coronary intervention) may be diagnostic
- NSTEMI
 - Partial occlusion leads to ischemia and damages heart muscle
 - Labs: elevated biomarkers
 - Requires PCI or thrombolytic
- STEMI
 - Full occlusion → infarction or death of heart muscle
 - Labs: elevated biomarkers
 - ST elevation on EKG
 - PCI needed

Angina

- Chest pain caused by reduced blood flow to heart
- NI for active angina: have pt rest, take vitals, ECG, give no more than 3 nitroglycerin (vasodilator) tablets 5 minutes apart, get help if no pain relief after med administration

Prinzmetal's Angina (aka variant, angina inversa)

- Chest pain that occurs at rest, usually at night or early morning
- Triggered by emotional stress, cold weather, smoking, medications that narrow blood vessels
- NI: Nitrates, calcium channel blockers

Myocardial Infarction

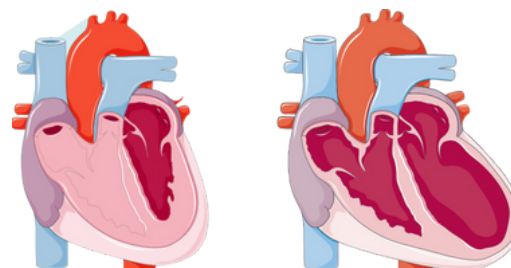
- Blockage of blood flow to heart; ECG shows ST elevation
- S/sx: severe chest pain/pressure, dyspnea, diaphoresis
- Women s/sx: neck, shoulder or jaw pain, fatigue, SOB, n/v, heartburn
- Meds to give: MONA
 - Morphine, oxygen, nitrates, aspirin
- Tests: Myoglobin, CK, and Troponin

Meds for MI

- M** Morphine
- O** Oxygen
- N** Nitroglycerine
- A** Aspirin

Heart Failure

- Heart is unable to pump enough blood to meet the body's oxygen demands
- Blood backs up into body (right sided HF) or into lungs (left sided HF)
- NI: O2 if needed, diuretics, monitor vital signs and I&O's, listen to heart and lung sounds, assess for hypoxia and edema, daily weights, fluid and sodium restriction



Normal

Heart Failure

Right-sided HF/ Cor Pulmonale

- Caused by left ventricular failure
- S/sx: peripheral edema, JVD, weight gain, hepatomegaly

Left Sided HF

- S/sx: pulmonary edema, dyspnea, orthopnea crackles, cough

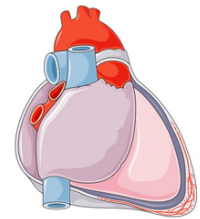
Cardiovascular

Cardiac Tamponade

- Fluid buildup in the pericardium → creates pressure → heart unable to pump effectively
- Medical emergency → cardiac arrest and circulatory shock
- Beck's Triad - hypotension, JVD, muffled heart sounds
- Pericardiocentesis needed to remove pericardial fluid

Pericarditis

- Inflammation of outer layer of the heart
- Indicator: presence of a friction rub. S/sx: pleuritic chest pain
- ECG finding: ST elevation and T-wave inversion
- NI: pain control, NSAID's, corticosteroids, monitor for s/sx of tamponade



Myocarditis

- Inflammation of the myocardium; typically caused by viral infection
- S/sx: fever, fatigue, chest pain, SOB
- NI: O2 as needed, bed rest, diuretics, ACEi, sodium restriction (if heart failure develops)

Endocarditis

- Inflammation of inner lining of heart
- Causes: Infection (TB or Staph), autoimmune conditions (lupus)
- S/sx: SOB, fever, palpitations, night sweats, chest pain
- NI: IV abx, promote oral hygiene, assess for signs of emboli, surgery if needed

Peripheral Arterial Disease (PAD)

- Chronic atherosclerotic disease caused by buildup of plaque within the arteries
- Commonly affects lower extremities → tissue necrosis (gangrene)
- Management: lower extremities below heart, moderate exercise to promote circulation, daily skin care, maintain mild warmth (socks, blankets), stop smoking, avoid stress and tight clothing, take prescription meds (vasodilators, antiplatelets) to increase blood flow

Coronary Artery Disease (CAD)

- Plaque buildup causes coronary arteries to narrow → limits blood flow to heart
- S/sx: angina, fatigue, SOB, heart attack
- NI: control BP, low fat and low salt diet, quit smoking, exercise, encourage healthy lifestyle

Scleroderma

- Causes abnormal blood flow in response to cold → Raynaud phenomenon

Fat Embolism

- Usually occurs after fractures of long bones
- At risk: femur fracture
- S/sx: SOB, confusion, tachycardia, lethargy
- Diagnostic/labs: "snow storm" on chest x-ray, increased ESR (detects inflammation in the body), hypocalcemia, respiratory alkalosis
- NI: Oxygen, IV fluids, bed rest, assess respiratory status

Deep Vein Thrombosis (DVT)

- Thrombus forms in vein; typically legs
- Risk → Virchow's Triad
- S/sx: warm skin, calf pain, edema
- NI: bed rest (prevent clot dislodgment), elevate extremity, compression stockings, administer anticoagulants and/or thrombolytics, measure circumference of affected area

Virchow's Triad

- S** Stasis of venous circulation
- H** Hypercoagulability
- E** Endothelial damage

Cardiovascular

Additional Notes:

- Improve perfusion → Elevate Veins, dAngle Arteries
- Assess BP in both arms of patients who have new diagnosed hypertension
- If pt has mechanical valves → need anticoagulation therapy for life to prevent thromboembolism
- If your patient has fluid retention, think heart issue first
- BNP (lab) is made by the heart and is released when the heart is under stress, trying to meet the demands of the body
 - BNP >100 pg/mL = heart failure

Neurology

Decorticate positioning

- Arms bent, legs straight, stiff
- Damage to cortex

Decerebrate positioning

- Arms and legs straight
- Severe damage to cerebellum or brain stem

Seizures

- Tonic- stiff then loss of consciousness
- Clonic- body spasms
- Myoclonic - quick muscle jerk
- Atonic - sudden lack of muscle strength
- Postictal phase - occurs after a seizure and ends when pt is back to baseline

Status epilepticus

- Seizure lasting more than 5 min or having multiple seizures without regaining full consciousness
- Convulsive vs nonconvulsive
 - Convulsive - most dangerous; tonic-clonic seizures

Seizure Nursing Interventions (NI):

- help pt lay down, place on side (to maintain airway), loosen tight clothes, give O2, record seizure duration

Strokes

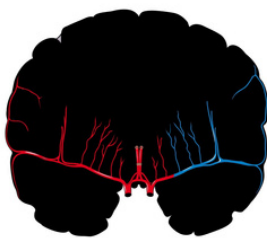
- Risk factors: hypertension, atherosclerosis, history of strokes, diabetes, smoking
- S/sx: motor loss (hemiparesis or hemiplegia), communication loss (dysphasia, dysarthria, apraxia, aphasia), vision changes, decreased mental acuity
- Start rehabilitation as soon as patient is stable
- NI: control BP, neuro assessment, position pt to decrease edema, bed rest, stool softeners

Ischemic Stroke; TIA

- Blockage of blood flow to brain due to clot
- NI: give TPA within 3-4 hours of symptoms

Hemorrhagic Stroke

- Brain bleed due to ruptured blood vessels
- Anticoagulation therapy is contraindicated



Aneurysm

- Buldge in a blood vessel that can lead to rupture → hemorrhagic stroke
- S/sx: vision changes, headache, nuchal rigidity, dizziness
- NI: promote calm environment, bed rest

Parkinson's Disease

- Central nervous system disorder that affects motor ability due to low dopamine levels
- S/sx: tremors, pill rolling movement, rigidity, stooped posture, bradykinesia, difficulties with gait
- NI: promote pt safety, schedule activities later in the day (conserves pt energy to perform self-care activities), calm environment, soft diet, physical therapy

Key Words

- Aphasia- inability to speak
- Dysphasia- difficulty speaking
- Dysarthria- slurred speech
- Apraxia- inability to perform movements or tasks

Neurology

Increased ICP

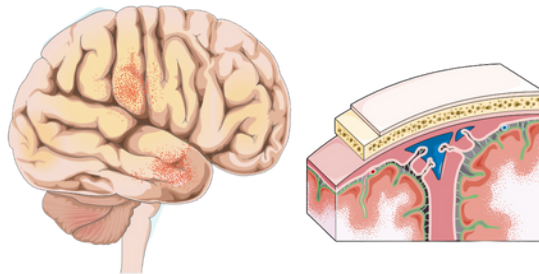
- Pressure builds in the skull; blocking brain circulation
- Sign = Cushing's Triad, change in LOC or pupil size, headache, blurry vision, vomiting
- NI: elevate HOB 30 degrees, keep body midline, promote calm environment, give stool softeners, tell pt to avoid Valsalva maneuver

Cushing's Triad

Bradycardia
Hypertension
Wide pulse pressure

Meningitis

- Inflammation of the membranes surrounding the brain and spinal cord
- Caused by viral, bacterial, or fungal infections
- Early warning signs: photophobia, drowsiness, confusion
- S/sx: nuchal rigidity (stiff neck), Brudzinski's (when pt neck is flexed → flexion of knees & hips) and Kernig's sign (pt unable to straighten leg when hip is flexed), fever, headache, muscle pain
- Diagnostic test: Cerebrospinal fluid test
 - If positive, CSF will have high protein and low glucose
- NI: give abx, Droplet/contact precautions, neuro assessment, vital signs, keep environment dark and calm, seizure precautions



Guillain Barre

- Immune system attacks it's own nerves, causing ascending paralysis
- S/sx: paresthesia (numbness and tingling), difficulty breathing, pain, vision changes
- NI: Assess respiratory status, mechanical ventilation may be needed

Multiple Sclerosis

- Chronic disease that affects the brain and spinal cord; more common in women
- Immune system attacks myelin sheath → breakdown of communication between neurons
- Bladder and bowel dysfunction occurs in most cases
- S/sx: numbness, vision problems, slurred speech, fatigue
- NI: promote tolerable exercises, implement rest breaks, create voiding schedule, high fiber diet and fluid intake

Myasthenia Gravis

- Muscle weakness and fatigue of voluntary muscles; d/t issues concerning ACh
- Improves with rest and worsens with physical activity
- S/sx: diplopia (double vision), ptosis (eyelid drooping), weakness, breathing issues
- NI: assess respiratory status, have trach kit at bedside, administer cholinergic meds, cluster care (to conserve pt energy), encourage deep breathing and coughing

Myasthenia Crisis

- Medical emergency due to worsening muscle weakness, causing respiratory failure
- Caused by undermedication, stress, or infection
- S/sx: worsening MG symptoms
- Diagnostic: positive Tensilon test

Cholinergic Crisis

- Occurs when there is excessive acetylcholine (ACh) in the neuromuscular junction
- Typically caused by too much anticholinesterase medication (given in myasthenia gravis)
- S/sx: cramps, diaphoresis, diarrhea
- NI: give antidote = atropine sulfate

Neurology

Amyotrophic Lateral Sclerosis (ALS)

- Progressive muscular disease caused by the degeneration of nerve cells
- Affects voluntary muscle movements, such as walking, breathing, and talking
- NI: assess respiratory status, encourage PT/OT/SP

Autonomic Dysreflexia

- Medical emergency seen in spinal cord injuries - T6 or higher
- S/sx: severe hypertension, bradycardia, sweating, anxiety, headache
- Nursing: elevate HOB to 90 degrees, remove tight clothing, administer antihypertensive medications

Basilar Skull Fracture

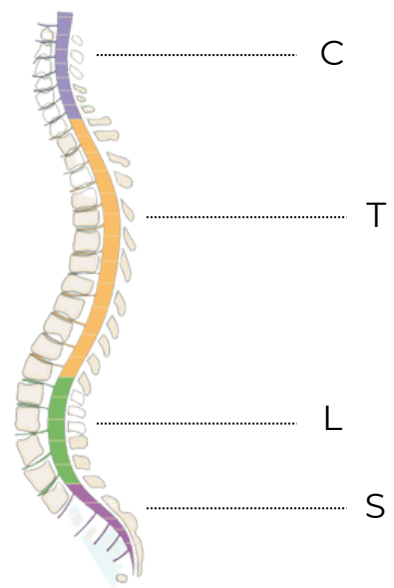
- Fracture of one of the bones at the base of the skull
- S/sx: Bruise behind ear (battle sign), periorbital hematoma (raccoon eyes), halo sign, hearing loss
- NI: assess ABC, neuro exam, GCS, CT scan

Wernicke's Encephalopathy

- Neuro disorder characterized by low thiamine (Vitamin B1)
- D/t alcohol abuse, eating disorders, or chemo
- S/sx: confusion, ataxia, abnormal eye movements

Spinal Cord Injuries

- If injury is between:
 - T1 - L4 - Paraplegia
 - C1 - C8 = Quadriplegia
 - C2 - C3 = Typically fatal
 - C4 and above = require ventilator
- NI: assess respiratory status, neuro checks, immobilize pt on backboard, keep body midline



Immunology

Immunoglobulins

- IgA- viral protection
- IgE- allergy and parasite infestation
- IgG- second antibody protection
- IgM- primary antibody protection

Allergies

- Basophils release histamine during an allergic response
- In pts with latex allergies → assess for allergies to bananas, kiwis, apricots, avocados, grape

Sepsis

- Medical emergency in which the body's response to an infection gets out of control, which can cause tissue damage and organ failure
- S/sx: Hypotension, tachycardia, SOB, fever, confusion

Anaphylaxis

- Severe allergic reaction that requires immediate intervention
- S/sx: nausea/vomiting, SOB, rash, anxiety

HIV

- Spread by contact through blood, semen, vaginal secretions, breast milk
- No cure but can be controlled. Pt is on standard precautions
- Priority: prevent infection
- Lab testing: positive ELISA, Western blot test, PCR
- If not treated, it can lead to AIDS

AIDS

- Virus caused by aids
- S/sx: low WBC, low CD4, low platelets, weight loss, fever, night sweats, weakness, infections
- Pt must maintain strict adherence to antiretroviral therapy
- Official diagnosis: CD4+ falls below 200 cells/mcl
- Incubation is variable, but typically occurs within 10 years after infection

Systemic Lupus Erythematosus (SLE)

- Autoimmune disease that causes extensive inflammation and tissue damage
- S/sx: butterfly rash, joint pain, fever, fatigue, weight loss, nephritis
- Triggers: sunlight, stress, drugs, pregnancy
- NI: teach patient to avoid sunlight exposure, provide steroids, advise patient to wash skin with mild soap

Oncology & Hematology

Radiation

- Monitor pt for infection due to leukopenia

Metastasis

- Common sites: liver, brain, bone, lung, lymph

Breast Cancer

- Modifiable risk factors: alcohol consumption, smoking, sedentary lifestyle, poor diet high in fat
- Non-modifiable risk factors: BRCA1 & BRCA2 mutations, having a 1st degree relative with breast cancer, history of uterine cancer, menarche before 12 years old, menopause after 55 years old
- Exam: check for lumps or hard knots in the shower, under arms and around entire breast area. Examine any changes in the shape of the breast and nipple by looking in the mirror.
 - Perform exam monthly, preferably post menstrual bleeding. If postmenopausal, exam should be done the same date every month.
- Mammography is helpful in early detection
- Treatment depends on stage of disease: mastectomy, chemotherapy, radiation, hormonal therapy

Post Mastectomy

- Elevate arm to avoid lymphedema, avoid heavy activity and lifting, do not wear constrictive clothing or get BP reading from affected arm

Testicular Cancer

- Small, hard lump on the front or side of testicle
- At risk: men whose testes have not dropped into the scrotum or whose testes dropped after age 6
- All men 14 years and older should do self exams monthly in the shower

Multiple Myeloma

- Cancer of plasma cells (WBC's that produce antibodies)
- Myeloma cells prevent the normal production of antibodies → infection

Radioactive Iodine

- ***FLUSH with at least 3-4 liters/day for 2 days and flush the toilet twice
- Limit contact with others to 30 minutes/day
- No pregnant nurses, visitors, and no kids

Immunosuppression

- Most oncologic meds cause immunosuppression → prevention of infection is vital
- Place pt in private room. Pt should NOT eat raw fruits or vegetables

Tumor Lysis Syndrome

- Tumor cells release their contents into the bloodstream
- Causes hyperkalemia requires insulin to IV solution to decrease potassium level. *Check blood glucose

Thrombocytopenia

- Place on bleeding precautions
 - Give soft bristled toothbrush
 - Decrease IM meds and do not insert anything (enema or suppositories)

Iron Deficiency Anemia

- Give iron PO with vitamin C or on an empty stomach
- Iron given IM- Infeon via Ztrack

Pernicious Anemia

- Needs to take Vitamin B12 for LIFE- due to lack of intrinsic factor in the gastric mucosa
- S/sx: tachycardia, pallor, beefy red tongue
- Shilling Test- determines how well one absorbs Vit. B12

*If a patient has low hematocrit and/or hemoglobin, assess for signs of bleeding (i.e. dark stools)

Nephrology

Acute Kidney Injury

- Kidney's unable to filter waste from the blood; potentially reversible
- S/sx: High level of creatinine and/or reduction in urine output, fluid retention (edema, HTN), changes in LOC, uremia
- NI: Monitor I&O, daily weight, BP, hyperkalemia, limit sodium and fluid intake, dialysis may be needed

Chronic Renal Failure

- Irreversible damage which causes uremia
- Dialysis needed to remove waste from blood
- Pt at risk for hyperkalemia and fluid overload
- NI: Restrict protein in CRF patients, monitor for fluid overload (JVD), daily weights, monitor I&O

Glomerular Filtration Rate (GFR) Ranges

- >90 = normal
- 60-89 = mild CKD; Stage 2
- 30-59 = moderate CKD; Stage 3
- 15-29 = severe CKD; Stage 4
- <15 = end stage renal disease ; Stage 5

Urinary Tract Infection (UTI)

- Upper - pyelonephritis; potentially life threatening → urosepsis
 - S/sx: Flank and back pain, fever, n/v
- Lower - urethritis & cystitis; commonly caused by E. coli
- Risk: pregnant women, pts w/ diabetes, men with prostatic hypertrophy, older adults
- NI: abx, teach pt to take entire course of abx, increase fluid intake, I&O's, encourage pt to void q2-3h, monitor for confusion in elderly pts, avoid catheterization

Urinary Tract Obstruction

- Blockage in one or both ureters, inhibiting urine to drain from kidneys to bladder
- Causes: calculi, strictures, tumors
- Location of stone can be determine by location of pain
- NI: pain control, increase fluid intake (to dislodge stone), IV abx (as needed), assess urine for stone, I&O's

Neurogenic Bladder

- Lack of bladder control due to CNS issues
- Also seen in pts w/ Parkinson's, MS, and spina bifida
- S/sx: UTI, urinary incontinence, kidney stones
- NI: create voiding schedule, double voiding, pelvic floor exercises, intermittent cath, pt may need indwelling cath

Benign Prostatic Hyperplasia (BPH)

- Enlargement of the prostate that occurs in older men
- S/sx: increased frequency or urgency to urine, nocturia, difficulty initiating urinating
- NI: encourage frequent voiding, increase fluid intake, monitor vital signs, I&O's, medications (antispasmodics, alpha-adrenergic antagonists)
- Transurethral Resection of Prostate (TURP) may be needed
 - Continuous bladder irrigation is needed post op to remove blood clots. Irrigate until urine is light pink.

Nephrotic Syndrome

- Leads to proteinuria → hypoalbuminemia → fluid shifts → edema

Nephrology

Additional Notes:

- Kidneys normally excrete 1mL/kg/hr.
- Typical adult urine output: 1500 - 2000 mL (varies based on condition)
- Indicator for renal status and fluid retention → body weight (weigh pt at same time & same scale daily)
- If pt is receiving dialysis → monitor for s/sx of shock (d/t blood loss)
- Restrict protein in CRF pts
- GFR is used as an indicator for protein consumption

Endocrine

Diabetes

Type I DM

- Insulin deficiency due to destruction of beta cells
- Without insulin, fats are metabolized for energy, which results in ketones acidosis

Hyperglycemia

hot and dry

Hypoglycemia

cold and clammy

Diabetic Ketoacidosis

- Sudden onset, increased ketones and acetone due to inadequate insulin dose
- Glucose >300, ketones in urine, fruity breath, confusion, Kussmaul's respirations, increased thirst and urination
- 1st thing to do → start IV infusion of normal saline
 - Severe dehydration occurs and must be rehydrated before insulin is administered
 - *Expect potassium to drop rapidly, so be ready with potassium replacement
 - Dropping glucose down too fast can cause increased intracranial pressure d/t water being pulled into the CSF

Type II DM

- Resistance or lack of insulin

Hyperglycemic Hyperosmolar Nonketonic Syndrome (HHNS)

- Glucose >600
- Severe dehydration, low potassium due to diuresis, no ketones or acidosis
- S/sx: altered LOC, increased thirst and urination, lethargy, coma

HbA1C

- Test that averages blood glucose levels over the past 90-120 days
- 4-6 = blood sugar level around 70-110
- 7 is ideal for a diabetic

Drawing up insulin: Nicole Richie RN

- Air into NPH, then air into Regular, draw up Regular insulin then NPH

Insulin

Type	Onset	Peak	Duration
Rapid (Aspart, Lispro, Glulisine)	<15 min	1-2 hours	3-6 hours
Short (Regular)	30-60 min	2-4 hours	6-10 hours
Intermediate (NPH)	2-4 hours	4-8 hours	10-18 hours
Long (Detemir, Glargine)	1-2 hours	No peak	18-24 hours

Dawn Phenomenon

- Early morning hyperglycemia in pts with diabetes

Somogyi Effect

- Occurs when pt takes insulin before bed, has hypoglycemia around 2-3am, and wakes up with hyperglycemia.
- The insulin lowers the blood sugar too much, causing a rebound effect

Endocrine

Additional Notes

- Fluids are the most important intervention in DKA and HHNS
- Do NOT give oral hypoglycemic meds to unconscious patients d/t risk of aspiration
- Additional insulin may be needed for patients on steroids (i.e. prednisone). Steroids increase glucose levels.

Diabetic neuropathy

- Earliest sign is microalbuminuria

Diabetes Insipidus: disorder of salt and water metabolism

- Excessive thirst and urination, dehydration, weakness
- Decreased ADH
- NI: monitor electrolytes, I&O's, give fluids, monitor urine specific gravity and serum osmolality

SIADH: water retention and dilutional hyponatremia

- Body retains water which can lead to water intoxication and hyponatremia
- Increased ADH
- Nursing interventions: daily weights (monitor for fluid overload), give diuretics, assess deep tendon reflexes, assess for neurological changes if sodium <115, maintain quiet environment and prevent overstimulation (prevent seizures)

Risks for metabolic syndrome

- Abdominal obesity, sedentary lifestyle, high levels of triglycerides
 - These increase the risk for coronary artery disease, DM Type II, stroke

Addison's Disease - hypocortisolism

- Dark pigmentation, weight loss, alopecia, decreased resistance to stress, GI distress
- Labs: hyperkalemia, hyponatremia, hypoglycemia
- *Most important assessment in Addison's → blood pressure, as it causes severe hypotension
- Help patient work on managing stress to avoid Addisonian crisis
- Addisonian crisis s/sx: confusion, weakness, nausea/vomiting, abdominal pain, hypotension, hypoglycemia, dehydration

Cushing's Syndrome - hypercortisolism

- Moonface/buffalo hump, hirsutism, hypertension, muscle wasting, weakness
- Hyperglycemia, hypernatremia, hypokalemia
- NI: monitor weight, I&O's

Grave's Disease - Hyperthyroidism

- Accelerates metabolism, causing weight loss and arrhythmias
- Fine/soft hair, sensitive to heat, irritability, exophthalmos
- Thyroid Storm: think HOT (hyperthermia), hypertension, tachycardia

Myxedema - Hypothyroidism

- Slows metabolism, dry skin and hair, sensitive to cold, fatigue
- Myxedema coma - due to severe hypothyroidism; medical emergency
- S/sx: decreased mental status, hypothermia, slows function of organs

Post-thyroidectomy

- Assess for hypocortisolism and temporary diabetes insipidus
- Patient position: semi-fowlers, prevent neck flexion, and have trach at bedside

Hyper-parathyroid

- fatigue, memory loss, body aches, difficulty sleeping, joint pain (d/t increased calcium), renal calculi
- Give calcitonin (regulates calcium levels in blood by decreasing it)
- Diet focused on low calcium and high phosphorus foods

Endocrine

Hypo-parathyroid

- CATS: convulsions, arrhythmias, tetany, spasms, stridor. Decreased calcium levels.
- Diet focused on high calcium and low phosphorus foods. Give Vitamin D to help calcium absorption

Chvostek sign

- Sign of hypocalcemia
- Twitching of facial muscles after tapping on a pts cheek

Trousseau sign

- Sign of hypocalcemia
- Carpopedal spasm (involuntary flexion of muscles in hand and wrist) resulting from ischemia
- How to test: place BP cuff on patients arm and begin to inflate (spasm is induced by compression)

Pheochromocytoma

- Hormone secreting tumor that can occur in the adrenal glands
- Causes hypertension, headache, tachycardia, hyperglycemia, diaphoresis
- Avoid stress and stimulating foods. Surgery to remove the tumor is usually required.

Gastrointestinal

Bowel Obstruction

- 2 types: mechanical (caused by an object) or non-mechanical (paralytic ileus)

Small Bowel Obstruction

- Causes: hernia, abd surgery, adhesions
- S/sx: cramping, n/v, pain is mid-abdominal, severe constipation
- NI: NPO, IV fluids, I&O's, NG-tube w/ low suction, pain management

Large Bowel Obstruction

- Medical emergency - risk for intestinal rupture
- S/sx: lower abdominal pain, n/v, diarrhea, constipation, no appetite
- NI: NPO, IV fluids, I&O's, NG-tube w/ low suction, pain management

Colostomy

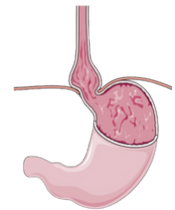
- Operation where a pt's colon is diverted to an opening in the abdominal wall; temporary or permanent
- NI: assess skin integrity around stoma, clean stoma with warm water and pat dry, prevent liquid stool in bag from overflowing, teach colostomy care

Dumping Syndrome

- Can develop after stomach surgery and causes abdominal pain
- Increase consumption of fats and proteins, eat small meals, lie down after meals, wait 1 hr after meals to drink liquids

Hiatal Hernia

- Part of stomach bulges through the diaphragm, causing heartburn and abd pain
- NI: eat small frequent meals, eliminate foods that worsen symptoms, have pt sit upright for one hour after eating, place pt in Fowler or semi-Fowler position (to reduce regurgitation)



Hiatal Hernia

Peptic Ulcer Disease (PUD)

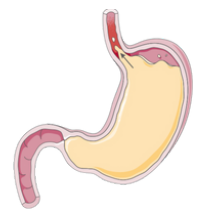
- Sore in the mucosal wall of the GI tract; can be caused by H. pylori
- Coffee brown emesis
- Duodenal → melana stool
- Gastric → hematemesis
- NI: Monitor stools and test for occult blood, give meds and mucosal healing agents, give small frequent meals

Gastroesophageal Reflux Disease (GERD)

- Regurgitation of stomach acid into the esophagus, causing heartburn, dysphagia, chest pain
- Pain typically occurs after eating and worsens when lying down
- Pt should avoid triggers: spicy food, caffeine, chocolate, citrus fruits
- NI: encourage small frequent meals, teach pt to stop eating 3 hours before bed, give H2 antagonists, PPI, antacids



Closed sphincter



Open sphincter

Esophageal Varices

- Enlarged veins in esophagus; occurs when blood flow to liver is blocked by the portal vein
- NI: maintain open airway, monitor for rupture and hemorrhage, suction if needed
- Management: insertion of esophagogastric balloon tamponade, coagulation factors, vitamin K, vasopressors

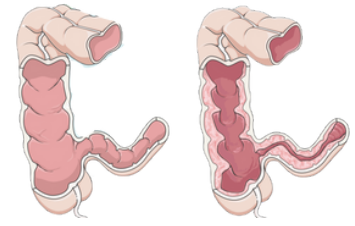
Stoma

- Opening on the abdomen; should be pinkish
- Dark/Dusky stoma = poor blood supply
- Black = necrosis
- Protruding = prolapsed

Gastrointestinal

Crohn's Disease

- Inflammation of digestive tract causing abdominal pain, diarrhea, malnutrition, weight loss
- Diet should be low fat, high protein, no dairy or spicy foods
- NI: monitor I&O and electrolytes, track weights, assess bowel patterns, educate pt to avoid alcohol, smoking, and caffeine

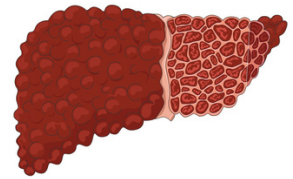


Ulcerative Colitis

- Inflammation of large intestine and rectum that causes liquid stools (blood & mucus), abdominal pain, rectal bleeding, anemia, fatigue
- NI: refer to Crohn's

Cirrhosis

- Liver damage which causes scarring and liver failure; causes jaundice and palpable liver
- Labs: elevated ammonia
- Slows metabolism of drugs, so they linger in the body
- Causes: hepatitis and chronic alcohol use
- NI: give ADEK vitamins, assess mental status, avoid anything that could cause bleeding, give lactulose (excretes ammonia), daily weights, monitor I&O, monitor protein intake



Hepatitis

Hepatitis A	*Ends in a VOWEL, comes from the BOWEL*
Hepatitis B	Blood and bodily fluids
Hepatitis C	Blood • More likely to turn into a chronic condition than B

Pancreatitis

- Caused by digestive juices or enzymes attacking the pancreas
- Signs - Cullen and Turner's sign
- Cullen sign - discoloration of umbilical area, Turner's sign - bruising of the flanks
- NI: Place patient in knee-chest position (decrease pain), NPO, typically will be placed on gut rest and need a PICC for TPN/lipids

Cholecystitis

- Gallbladder inflammation d/t the buildup of bile in the gallbladder
- Causes cholelithiasis (gallstones), which blocks bile from being excreted
- S/sx: n/v, epigastric pain that radiates to right scapula after eating high fat foods, RUQ pain, rebound tenderness, jaundice, Murphy's sign, guarding
- NI: NPO, IV fluid, abx, anticholinergics, pain control

Enema

- Monitor for water intoxication (signs/symptoms: weakness, pallor, dizziness, diaphoresis)

Additional Notes:

- Stopping TPN abruptly → hypoglycemia
- Injured hepatic cells release AST and ALT. High levels of AST & ALT = liver injury or disease

Musculoskeletal / Ortho

Ascending Stairs

- Step with stronger leg, move the cane to the next step, then move the weaker leg

Descending Stairs

- Lead with the cane, bring weaker leg down, then step down with stronger leg

*Remember: Up with the good, down with the bad

5 P's of fractures

- Pain, pallor, pulselessness, polar (cold), paresthesia

Types of Fractures

Displaced - bone breaks in pieces and moves out of alignment

Non-Displaced - bone breaks and stays in alignment

Open - bone breaks through skin

Closed - skin is not broken

Comminuted - bone breaks into multiple pieces

Complete - bone breaks across entire section

Incomplete - bone does not break across entire section

Oblique - bone broken at angle

Compression - bones are crushed

Transverse - bone breaks in a straight line

Segmental - bone broken in at least 2 places; a segment of bone is separated by breaks

Stress - hairline fracture

Spiral - fracture spirals around bone; caused by twisting injuries

Greenstick fracture - bone breaks on one side and bends on the other; common in peds

Buck's Traction

- Ensures proper alignment, weights hand freely and do NOT touch the floor
- Do not remove or lift weights without a doctor's order

Casts

- NEVER insert anything under cast; allow 24-72 hrs for cast to dry
- To relieve itch, use cool air from blow dryer

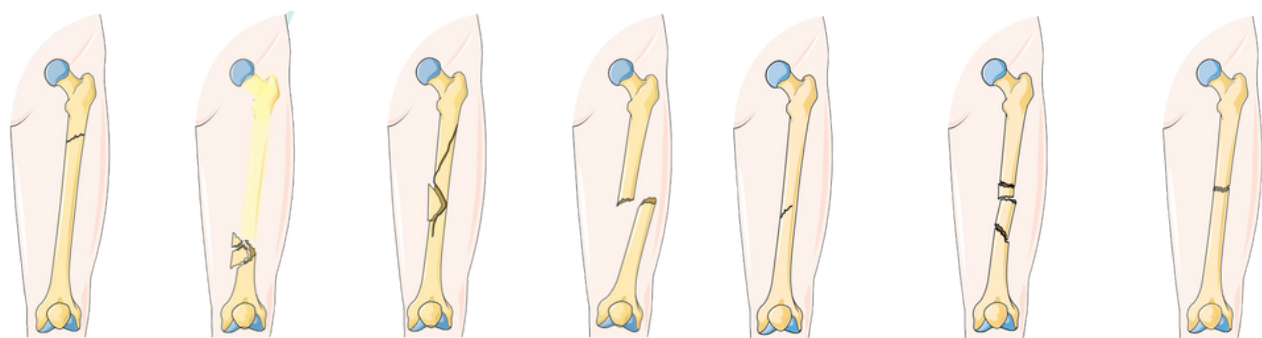
Hip Fracture

- Monitor for s/sx of hemorrhage
- Thromboembolism is the most common complication; prevention includes passive ROM exercises, elevate foot of bed to 25 degrees, wear elastic stockings
- NI: avoid internal and external rotation of more than 60 degrees, avoid weight bearing on affected leg, leg position should be ABDUCTION (prevents dislocation), avoid crossing legs, monitor neurovascular status of the injury site (skin color, sensation, cap refill, temp, mobility)

Cane Walking

- C** Cane
- O** Opposite
- A** Affected
- L** Leg

Cane always moves before weaker leg



Oblique

Comminuted

Spiral

Complete

Greenstick

Segmental

Transverse

Musculoskeletal / Ortho

Joint replacement

- Major complication post op is infection

Immobility d/t orthopedic limitations

- Pt's are prone to venous thrombosis, skin integrity issues, and buildup of urinary calculi

Osteoporosis

- Bones become weak and brittle; pts are prone to fractures
- S/sx: hump or kyphosis, back pain, loss of height
- At risk: postmenopausal, thin, white women
- NI: encourage a high calcium diet, weight-bearing exercise, supplemental calcium (& Vit D for absorption)

Osteomyelitis

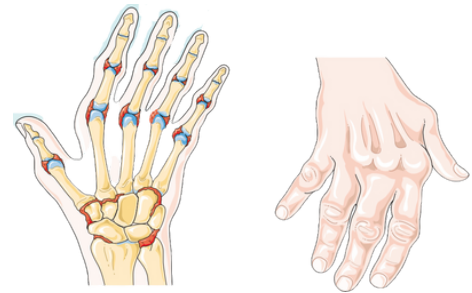
- Infectious bone disease
- NI: Get blood cultures and antibiotics. If necessary, surgery may be required to drain abscess

Gout

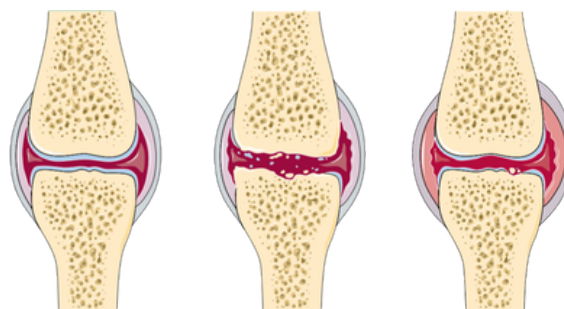
- High uric acid in the body causes buildup of uric acid crystals in the joints
- Causes severe pain, inflammation, and redness
- NI: low purine diet (avoid sugar, organ meats, alcohol, yeast), promote high fluid intake, NSAID's, corticosteroids, rest as needed

Rheumatoid Arthritis

- Autoimmune and inflammatory disease that affects joints
- Pain relief is high priority
- During an acute episode, do NOT assess range of motion, as this is painful. Teach patient not to exercise swollen joints
- S/sx: generalized weakness, fatigue, stiffness, joint deformity
- NI: Assess the pt using strength testing, palpation, and inspection. avoid weight bearing on inflamed joints, rest, PT & OT
- Treatment: corticosteroids, immobilization, splinting, immobilization, NSAID's (pain), rest
- Diagnostic: Rheumatoid factor blood test



Rheumatoid Arthritis



Normal

Osteoarthritis

Rheumatoid arthritis

Integumentary / ENT / Eyes

Wound Drainage

Serous	Clear, watery
Fibrinous	Tan to yellow, may be dry or slightly moist
Serosanguinous	Pink, watery
Sanguinos	Thick red blood, deep wound 
Purulent	Thick liquid; yellow, green, gray Sign of infection
Hemorrhagic	Bright red, damage to artery or vein

Psoriasis

- Chronic skin disease that causes a rash with scaly patches; itchy
- Meds: topical corticosteroids, methotrexate
- NI: Educate pt on triggers - stress, dry skin, trauma, infection. Advise patient to limit or eliminate alcohol and smoking.

Shingles (Herpes Zoster)

- Painful rash caused by reactivation of chickenpox virus
- S/sx: fluid-filled blisters, fever, shooting pain, pruritis
- Treatment: pain relief and antiviral medications
- Place on contact precautions

MRSA

- Resistant bacteria that is difficult to treat with antibiotics
- Causes painful red bumps on skin that may turn into abscesses
- Place on contact precautions

Meniere's Disease

- Disorder of the inner ear that causes vertigo and hearing loss
- Nursing: administer diuretics, restrict sodium, lay on affected ear when in bed

Eyes

- OS - Left eye
- OD - Right eye
- OU - Both eyes

Strabismus (crossed eyes)

- Can be treated with botox
- Patch the GOOD eye so the weaker one gets stronger

Glaucoma

- In angle closure glaucoma do not take meds that dilate the pupil → can precipitate acute and severely increase intraocular pressure
 - NO atropine medications

Critical Care

Pulmonary Capillary Wedge Pressure (PCWP)

- Normal: 8-13 mm Hg
- Measures left atrial pressure

Cardiac Arrhythmias Treatments

- Uncontrolled atrial fibrillation → synchronized cardioversion
- Ventricular dysrhythmias → Lidocaine
- Ventricular fibrillation or unstable ventricular tachy → Defibrillate
- Sinus bradycardia → Atropine

Implantable Cardioverter Defibrillator (ICD)

- Given with life-threatening dysrhythmias
- Pacemaker capabilities
- Pt teaching: refrain from lifting affected arm above shoulder until physician says OK. Firing can be painful.

Ventilator Alarms

- High Alarm - obstruction due to increased secretions, tubing kink, or if they pt coughs or gags on tubing
- Low Alarm - disconnection or leak; pt stops breathing
- NI: assess what alarm went off and provide intervention as needed; assess oxygen level; call RT and/or MD if assistance is needed

Ventilator Alarms - HOLD

H	High alarm
O	Obstruction
L	Low Alarm
D	Disconnection

Positive End-Expiratory Pressure (PEEP)

- Used during mechanical ventilation to maintain airway pressure above atmospheric pressure
- Improves oxygenation while decreasing FiO₂ to a less toxic level

Tracheostomy

- Trach care: clean inner cannula, suction, and place new dressing
- Always have suction equipment at bedside

Blood Transfusion

1. Find pt's blood type and crossmatch with another RN
 - a. Need at least 2 pt identifiers
 - b. Blood must be administered within 20 minutes
2. Obtain baseline vital signs, educate pt on s/sx of transfusion reaction
3. Prepare Y tubing with NS; clamp NS
4. Spike blood bag and leave blood clamp open
5. Infuse slowly for first 15 minutes and stay with pt to assess for reactions (nausea, vomiting, chills, hypotension, fever, back pain, dyspnea)
6. Take vital signs after 15 minutes AND when infusion is done
7. Deliver blood over 2-4 hours
8. After infusion is complete, open NS clamp to flush blood from tubing

Reaction Suspected

- Immediately STOP transfusion
- Disconnect tubing and assess s/sx of reaction
- STAY with the pt. NEVER leave pt alone until MD comes or symptoms resolve.
- Notify MD and blood bank
- Document event

Critical Care

Shock

Shock

- Widespread reduction of tissue perfusion that can lead to organ damage (d/t lack of oxygen and nutrients)
- Early s/sx: restlessness and agitation (d/t cerebral hypoxia)
- S/sx: hypotension, tachycardia, weak peripheral pulses
- Shock can lead to multiple organ dysfunction syndrome (MODS), systemic inflammatory response syndrome (SIRS), respiratory distress, pleural effusion, death
- NI: Oxygen/ventilation, IV fluid resuscitation (restore tissue perfusion), assess vital signs, cardiac meds (varies based on type of shock)

Type of Shock	Cause	Result
Hypovolemic (most common)	Loss of fluid and/or blood	Hypotension, tachycardia, weak thready pulse, agitation, pale cool skin, decreased UO
Cardiogenic	Heart unable to pump enough blood to meet body's demand Ischemia or lack of tissue perfusion	Tachypnea, SOB, tachycardia, hypotension, weak thready pulse, pale cool skin, decreased UO Decreased cardiac output
Distributive (Vascular) 3 Types: Anaphylactic	Allergy reaction	Tachycardia, hypotension, dyspnea, cough, pruritis, restlessness, decreased LOC
Neurogenic	Spinal cord injury	Bradycardia, hypotension, warm dry skin, blue lips
Septic	Endotoxins from gram-negative bacteria	Tachycardia, hypotension, tachypnea, bounding pulses, cool pale extremities, decreased UO
Obstructive (Circulatory)	Obstruction prevents perfusion to organs Compartment syndrome, emboli, tamponade, tension pneumothorax	Tachycardia, hypotension, change in LOC, decreased UO, cool pale skin Decreased cardiac output

Disseminated Intravascular Coagulation

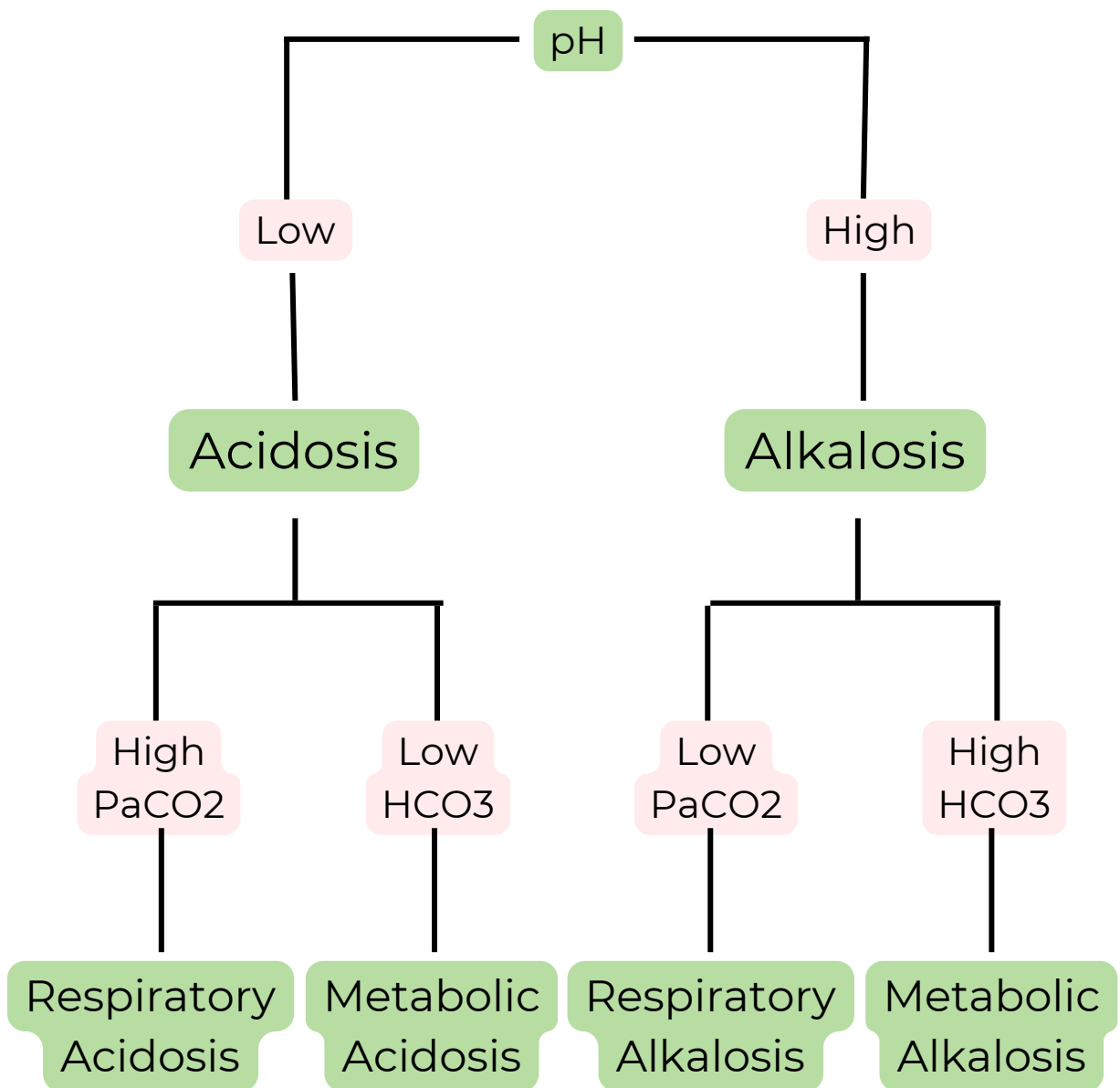
- Coagulation disorder that causes thrombosis and hemorrhage
- Labs: prolonged PT & PTT, decreased fibrinogen and platelet count
- S/sx: petechia, hemoptysis, oozing from IV sites & gums, GI bleeding, hypotension, tachycardia
- NI: assess for s/sx of bleeding, monitor PT/INR levels, IV heparin during first phase to inhibit coagulation, provide emotional support

Multi-Organ Dysfunction Syndrome (MODS)

- Inadequate tissue perfusion and increased oxygen demands

Critical Care

ABG



Pediatrics

3 H's of peds on why they code

- Hypoxia
- Hypothermia
- Hypoglycemia

Symptoms of hypoxia

- Difficulty eating, sternal retractions, nares flaring

Urinary output

- Infants and children should be 1-2 mL/kg/hr

Vaccines

- No LIVE vaccines for immunocompromised children
 - Varicella-zoster
 - Measles/mumps/rubella
 - Rotavirus
 - Yellow fever

Immunizations

- Ask if pt has an allergy to eggs before flu shot and MMR vaccine
- Child can still receive immunizations if they have a cold
- Children with HIV: avoid Polio and Varicella vaccinations (live). Can still get pneumococcal and influenza
 - Avoid MMR only if the child is severely immunocompromised

Varicella (Chickenpox)

- Viral disease that causes itchy skin lesions
- Airborne precautions; Spread by direct contact & droplet
- Meds: acyclovir, analgesic, antihistamine

Rubeola (Measles)

- Highly contagious viral disease characterized by rash, Koplik's spots (lesions in the mouth), photophobia
- Airborne precautions
- "10 day measles"
- NI: bed rest, encourage fluids, ensure pt has short nails (protect skin), antipyretics, keep skin moist with prescribed lotion

Rubella (German Measles)

- Viral disease characterized by rash that starts on face and spreads, low grade fever, sore throat, enlarged lymph nodes
- Droplet precautions
- "3 day measles"
- NI: keep infected pt away from pregnant women as they are a risk to the fetus

Pertussis (Whooping Cough)

- Infectious respiratory disease characterized by uncontrollable coughing, respiratory distress, cyanosis, exhaustion
- Droplet precautions
- NI: Abx, oxygen and suction as needed

Neutropenic Precautions

- Wash hands often
- Avoid contact with sick people, those recently vaccinated, and large crowds
- No milk, yogurt (live cultures), fresh fruit, or vegetables

Lead poisoning

- Assess if child lives in a home built before the 1950's
- Elevated blood lead levels >5 mcg/dL requires follow-up blood work
- Chelation therapy given if blood lead levels are high

Pediatrics

Ears

- Pull pinna **down and back** for kids < 3 years old when putting in eardrops

Otitis Media

- Infection of the middle ear
- To prevent: Feed upright, breastfeed for first 6 months, immunizations
- S/sx: fever, child may pull ear, enlarged lymph nodes, discharge from ear
- NI: abx, reduce body temp through tepid bath, position child on affected side, may need tympanostomy tubes placed

Lice

- S/sx: itchy scalp, irritability, red bumps on affected areas
- NI: Use PPE to assess scalp, behind ears, base of neck, and crown of head for lice, use Wood's lamp (black light) to find lice, apply pediculicide shampoo, comb hair with nit comb, change linen and clothes daily

Impetigo

- Red sores around nose and mouth caused by bacteria (typically staphylococcus). Sores rupture, ooze, then turn yellow-brown
- Risk: children aged 2-5, close contact with others, warm/humid weather
- NI: Contact precautions, keep sores open to air, abx (topic and oral), promote proper hygiene

Cleft Lip/Palate

- Pt born with opening in roof of mouth and/or lip due to failure of bone/soft tissue to fuse
- Cleft lip repair done at 3-6 months; cleft palate repair done 6-24 months
- Pt at risk for otitis media and speech impairment
- NI: promote family bonding and grieving, teach family surgery is available, assist with feeding using ESSR, bulb syringe at bedside to remove oral secretions

ESSR

- E** Enlarge Nipple
- S** Stimulate infant to suck
- S** Swallow
- R** Rest

Esophageal Atresia with Tracheoesophageal Fistula

- Congenital abnormality in which the esophagus does not fully develop
- 3 C's of TEF - coughing, choking, cyanosis; also causes excess salivation
- Surgical emergency
- NI: NPO, IV fluids, gastrostomy tube care, suction, give pacifier to promote developmental needs
- Post op- monitor for dysphagia, drooling, regurgitated food

Pyloric Stenosis

- Swelling of the pylorus (muscle b/w stomach and intestines) that occurs between birth - 6 months.
- Causes forceful vomiting → dehydration
- Treated with surgery - pyloromyotomy

Intussusception

- Part of intestine telescopes inside another, resulting in partial or complete obstruction
- S/sx: abd pain (raises legs up to abd), currant jelly stools, sausage shaped mass in RUQ, vomiting
- NI: assess for shock and perforation, IV fluids, barium enema, I&O's

Hirschsprung's Disease

- Absence of ganglion cells in colon and rectum → peristalsis → obstruction
- S/sx: no meconium within 24 hrs after birth, distended abd, constipation, ribbonlike stools (older children)
- Pre-op: bowel cleansing, measure abd girth, assess for perforation
- Post-op: Teach family temporary colostomy care

S/sx Perforation

- vomiting
- increased abd pain
- abd distention
- cyanosis
- dyspnea

Pediatrics

Reye Syndrome

- Disorder that causes severe encephalopathy and hepatic dysfunction
- Causes: pt recovering from viral infection (flu or chickenpox), aspirin use
- S/sx: lethargy, n/v, diarrhea, confusion
- NI: neuro assessment, give mannitol (if prescribed), I&O's

HIV

- Children born from HIV positive mothers should be given antiretroviral treatment
- Source of infection: perinatal transmission, breast milk, HIV infected blood products, sexual abuse

Sickle Cell Anemia

- Main interventions: fluids and pain relief

Leukemia

- Cancer of the blood or bone marrow
- May cause epistaxis (d/t low platelets), hyperplastic gums (overgrowth of gum tissue around teeth), increase in WBC count, weakness

Hemophilia

- X linked recessive disorder characterized by the inability to form clots
- S/sx: increased bleeding, bruising, epistaxis, blood in stool or urine
- NI: assess for bleeding, administer clotting factors

Von Willebrand disease

- Genetic bleeding disorder that inhibits coagulation d/t low levels of von Willebrand factor
- S/sx: bleeding of mucous membranes, epistaxis, heavy menstrual bleeding

Epiglottitis

- Medical emergency that causes airway obstruction
- Do NOT examine throat (can lead to spasm → obstruction)
- S/sx: fever, barking cough, dysphagia, drooling, restlessness, child assumes tripod position
- NI: maintain airway, do not leave child alone, keep pt in upright position, cool mist O2, prepare for intubation/trach, IV abx, do NOT measure oral temp

Tripod Position

Sitting upright with chin out, tongue protruding

Bronchiolitis

- Respiratory virus typically caused by RSV that is characterized by thick secretions
- S/sx: cough, rhinorrhea, congestion, tachypnea, wheezing
- Teach parents to use saline nose drops and suction nares with bulb syringe
- NI: Contact isolation, monitor respiratory status, suction airway with bulb syringe, O2 as needed, oral and IV fluids

Cystic Fibrosis

- Autosomal recessive disorder that causes secretions to be thicker and stickier
- CF severely damages the lungs, digestive system, and reproductive system d/t thick mucus causing obstruction
- Typically occurs in white infant or children
- S/sx: recurrent respiratory infx, pulmonary congestion meconium ileus at birth, steatorrhea (excessive fat, greasy stools), bad smelling bulky stools, skin that tastes salty
- NI: monitor respiratory status, IV abx, pancreatic enzymes (take with food), fat-soluble vitamins (ADEK), respiratory treatments: oxygen, nebulizer, vest, teach pt they will need a diet high in calories, protein, mod-high fat, low carbs, exercise

Pediatrics

Congestive Heart Failure (CHF)

- Heart is unable to meet the demand's of the body
- S/sx: SOB, tachypnea, cyanosis, weight gain & edema, difficulty feeding, hepatomegaly
- NI: daily weights, vital signs, elevate HOB, O2 as needed, diuretics and digoxin as prescribed, I&O's, low sodium and/or fluid restriction
- Weigh pt daily at the same time and on the same scale

Cyanotic Heart Disease

- R → L shunt
- Decreased blood flow
- Mixed blood is pumped into systemic circulation
- S/sx: cyanosis, squatting, syncope, clubbing

Creased Pulmonary Blood Flow:

Tetralogy of Fallot

Combination of 4 defects:

1. VSD
2. Pulmonary stenosis
3. Misplaced aorta
4. Right ventricle hypertrophy

Tet Spells - relieved by squatting or being in knee-chest position

Tet requires surgical intervention

Mixed Blood Flow:

Transportation of the Great Vessels

- Pulmonary artery leaves L ventricle, aorta exists from R ventricle

Truncus Arteriosus

- Aorta and pulmonary artery is connected
- Mixed blood from R and L ventricle causes cyanosis

Surgical intervention needed

Acyanotic Heart Disease

- L → R shunt
- Increased pulmonary blood flow or obstructive defects
- S/sx: murmur, fatigue, CHF, growth restriction

Increased Pulmonary Blood Flow:

Ventricular Septal Defect (VSD)

- Hole between ventricles

Atrial Septal Defect (ASD)

- Hole between atria

Patent Ductus Arteriosis (PDA)

- Opening between aorta and pulmonary artery
- Typically closes within 72 hr after birth
- Increased blood flow → pulmonary HTN

Obstruction of Blood Flow from Ventricles:

Coarctation of the Aorta

- Obstructive narrowing of aorta
- HTN found in upper extremities and decreased/absent pulses in lower extremities

Aortic Stenosis

- Obstructive narrowing before, at, or after aortic valve

Cyanotic
Unxygenated blood enters
systemic circulation

Acyanotic
Oxygenated blood enters systemic
circulation

Both acyanotic and cyanotic have abnormal circulation.

Kawasaki Disease

- Systemic vasculitis that casues damage to vessels, mucus membranes, lymph nodes, and skin
- S/sx: Strawberry tongue, rash on trunk and genitals, redness or peeling of hands/feet, high fever, n/v, joint pain
- NI: administer IVIG and aspirin, antipyretics, I&O's, mouth care, place cool compresses and lotion on skin

Rheumatic Fever

- Inflammatory disease that affects the tissues of the heart, blood vessels, joints, and skin
- Occurs weeks after untreated strep throat or scarlet fever
- Ask parents if the child was recently sick
- S/sx: SOB, chest pain, fever, migratory joint pain, chorea (irregular involuntary movements), rash, subcutaneous nodules over bony prominences
- NI: abx, aspirin (for antiinflammatory and anticoagulation), bed rest, vital signs

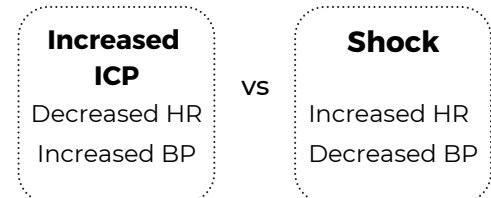
Pediatrics

Cerebral Palsy

- Irreversible disorder that causes neuromuscular issues of dyskinesia (involuntary movements) or spasticity
- S/sx: involuntary movements, poor sucking, abnormal posture, scissoring of legs, seizures, delayed developmental milestones
- NI: PT/OT/ST, place pt in upright position when eating (prevent aspiration), provide support to family, anticonvulsant meds, diazepam (muscle spasms)

Hydrocephalus

- Abnormal buildup of CSF within the ventricles in the brain → pressure on brain tissue
- Toddlers and children show signs of ICP: increased BP, decreased HR
 - Increased ICP is the opposite of shock
- Infants with increased ICP: enlarged head circumference, lethargy, irritability, bulging fontanelles, sunset eyes, widening suture lines
- S/sx: irritability, change in LOC, waking up w/headache, vomiting, unequal pupil size, seizures
- NI: assess pt's baseline data to compare new s/sx of ICP, seizure precautions, elevate HOB, prep for ventricular shunt placement (drains excess fluid)



Spina Bifida

- Neural tube defect that occurs when the spinal cord fails to form properly
- Prevention: teach women in childbearing years to take minimum 400 mcg folic acid daily
- S/sx: dimple at base of spine, presence of sac on lumbar, flaccid paralysis
- NI: screen pt for latex allergy
- Post-op: teach family and pt (if applicable) to self catheterization, give continence meds, high fiber diet, increased fluids, assess skin integrity, ROM exercises

Hypospadias

- Opening of urethra is located on the ventral side of the penis, not the tip
- S/sx: altered voiding stream, undescended testes, inguinal hernia
- NI: surgery needed, monitor urinary drainage post op, IV fluids, teach family home care for catheterization

Infant Feeding Regimen

- Breastfeed from birth - 6 months
- Introduce solid foods b/w 4-6 months
- Wait 5-7 days after introducing new foods to assess for allergic reactions
- Cow's milk ok to give at 1 year old

Nutrition Notes

- Do NOT give honey to pts under 1 years old. They are at risk for botulism
- Look out for questions regarding how much milk they are drinking (more than 3-4 cups/day). Too much milk reduces the amount of iron they intake so watch for anemia.
- Patients with PKU cannot break down phenylalanine. Avoid high protein foods - meat, dairy, nuts, legumes. Do NOT give them aspartame.

Additional Notes:

- Normal urine output for infants and children is 1-2 mL/kg/hr
- The usage of ipecac for poison removal is no longer recommended by the American Academy of Pediatrics. The nurse should teach parents to not induce vomiting, as it may be more harmful
- Do NOT use tongue blade during a seizure, as it can damage oral cavity

Maternity

Terminology

Gestation

- Time from fertilization until date of delivery

Gravida

- Pregnant woman

Gravidity

- # of pregnancies

Parity

- # of births past 20 weeks gestation (born alive or not)

Nullipara

- Woman who has not had a birth greater than 20 weeks gestation

Primipara

- Woman who has had 1 birth that happened after 20 weeks gestation

Nulligravida

- Woman who has never been pregnant

Primigravida

- Woman who is pregnant for the first time

Multigravida

- Woman who is pregnant for at least the 2nd time

GTPAL

- Gravidity
- Term births (at least 37 weeks)
- Preterm births
- Abortion/miscarriages
- Current living children

Pregnancy Length

- Length: 280 days, 40 weeks, 9 calendar months, 10 lunar months (28 day months)
- 1st Trimester: starting from the first day of last menstrual period - 13 weeks
- 2nd Trimester: 14 -26 weeks
- 3rd Trimester: 27 - 40 weeks

Pregnancy

- Causes a hypercoagulable state that protects the mother from hemorrhage after birth → may lead to thrombus formation

Nagele Rule for Estimating Date of Birth

- Find the first day of the last menstrual period (LMP)
- Add seven days
- Then subtract 3 months
- Ex: LMP is November 1, 2022
 - Add 7 days = November 8, 2022
 - Subtract 3 months = August 8, 2023

Fundal Height

- Measurement of the uterus to determine gestational age of fetus

Fetal Heart Rate

- Can be heard via Doppler at 10-12 weeks gestation

Prenatal Visits for Low-Risk Pregnant Women

- Visit MD every 4 weeks until 28 weeks, every 2 weeks from 28-36, then once a week from 36-40 weeks

Defect Screenings

- Some states require screening for neural tube defects
- They will use maternal alpha-fetoprotein (AFP) levels or amniotic fluid AFP levels. Note this specific test is prone to false positives/negatives

Maternity

Amniocentesis

- Done early in pregnancy → bladder must be full
- If done late in pregnancy → empty bladder (to avoid being punctured)

Epidural

- Hydrate before

C-section

- Increases risk for DVT (deep vein thrombosis)

Placenta abruption

- Rigid boardlike abdomen, dark red vaginal bleeding
- Severe pain
- Monitor I&O

Placenta Previa

- Soft abdomen
- Painless, bright red vaginal bleeding

Amniotic fluid

- Fluid is alkaline, which turns nitrazine paper blue. Urine and vaginal discharge are acidic and will turn it pink
- If the fluid is yellow with particles → meconium stained

Folic Acid

- Give before pregnant and throughout pregnancy
- Vitamin; used for anemia
- Leafy vegetables, eggs, citrus foods

NSAID's

- May take in 1st and 2nd trimester only if benefits outweigh the risks
- Avoid in 3rd trimester due to risk of premature closure of the ductus arteriosus in the fetus

Acetaminophen

- Pain reliever/fever reducer commonly taken during pregnancy
- Do not exceed 4g per day

Magnesium Sulfate

- Used in preeclampsia
- Therapeutic range: 4-8 mg/dL

Nonstress Test

- Noninvasive test in which fetal heart rate is monitored in response to fetal movement
- Mom pushes event marker every time she feels movement or when the FHR strip records movement
- Performed after 28 weeks gestation
- Reactive result = Normal
 - Fetal HR increased with movement (15 BPM for 15 seconds)
 - Indicates adequate blood flow and oxygen to fetus
- Nonreactive result = Further testing needed (Stress Test or BPP)
 - Fetal HR does not increase with movement
 - Other tests will determine if the result is due to sleep patterns, maternal medications, or fetal hypoxia
- If no fetal movement is shown, suspect fetus is sleeping
 - Stimulate fetus acoustically or have mom move fetus around

Maternity

Stress Test

- Induction of uterine contractions to show how the baby will respond during labor
- Contractions are induced by nipple stimulation or oxytocin infusion
- Performed after 34 weeks gestation and only if mother had an atypical result from a nonstress test or biophysical profile
- Negative result = Normal
 - Fetus HR does not slow during contraction → fetus properly responding to stress (3 contractions in 10 min)
- Positive result = Abnormal
 - Fetus HR slows during contraction → fetus is under stress

Biophysical Profile

- Nonstress test + ultrasound to assess the health of the fetus
- 5 areas are assessed:
 - Body movement, breathing movement, muscle tone, heartbeat, and amniotic fluid
 - Each area is scored as either 0 (abnormal) or 2 (normal)
 - Scores: 8-10 = normal, 6 = borderline, <6 = sign of possible complication

APGAR

Sign/Score	0	1	2
Appearance	Blue/pale	Body pink, extremities blue	Pink
Pulse Rate	None	< 100	> 100
Grimace	None	Grimace	Cough, Sneeze
Activity	Limp	Some	Active, flexed
Respiration	Absent	Slow/irregular	Strong cry

Scoring: 0-3 = Severely Depressed (Resuscitate), 4-6 = Moderately Depressed, 7-10 = Excellent

VEAL CHOP

FETAL HEART RATE PATTERN	ETIOLOGY
Variable decels	Cord compression
Early decels	Head compression
Accels	OK (no issue)
Late Decels	Placental insufficiency

NI: change pt position, administer O2, discontinue oxytocin (Pitocin), notify MD

Positioning

- Cord prolapse → knee-chest position or push the presenting part off the cord until a c-section is done
- Cord compression → Trendelenberg
 - Removes pressure from the cord
- Late decels → turn to left side (allows more blood flow to placenta)

L/S Ratio

- Test that examines lung maturity, lung surfactant, and development
- Most important determinant of fetal maturity for survival post birth

Station

- Location of presenting part in cm above or below the spine; the specific part of the baby that is leading the way through the birth canal (e.g. head, shoulder, feet, etc.)
- 0 - Engaged; at ischial spine
- Minus: above ischial spine (e.g. -2)
- Plus: below ischial spine (e.g. +2)

Cervical Effacement

- Ripening; cervix shortens, thins, and softens to prepare for birth

Leopold Maneuvers

- Palpation of the abdomen to determine fetal presentation, position, lie, and engagement

Maternity

Stages of Labor

1st Stage	Contraction & Cervical Dilation True onset of labor - complete dilation of cervix Longest Stage: 2-18+ hours 3 Phases • Latent : 0-4 cm cervical dilation ◦ Irregular, mild contractions • Active: 4-7 cm cervical dilation ◦ Contractions 5-8 min apart ◦ Moderate contractions; increased anxiety and discomfort • Transition: 7-10 cm cervical dilation ◦ Contractions 1-2 min apart ◦ Contractions severe, nausea, hiccups, irritability
2nd Stage	Pushing and Birth of Baby 10 cm dilated → delivery of baby
3rd Stage	Delivery of Placenta
4th Stage	Recovery & Baby Bonding • During the first 2 hours after birth, where breastfeeding is established

True Labor	False Labor
Lower back pain that radiates to abd Contractions that worsen with ambulation Cervical dilation and effacement	Abdominal discomfort Contractions decrease with ambulation No dilation or effacement

Hyperventilation during Labor

- Can cause respiratory alkalosis
- NI: give mom paper bag to breathe

Nitrazine Paper

- Used to determine if membrane has ruptured
- The paper turns dark blue/back

Oxytocin (Pitocin)

- Causes uterine contractions
- Give after placenta is delivered
 - Giving it before may can placenta retention → hemorrhage and infection

Lochia Types

- Rubra- bloody discharge that lasts 2-3 postpartum
- Serosa- pinkish brown discharge that lasts 1 week postpartum
- Alba - thick, white/yellow discharge that lasts up to 4 weeks postpartum

Lochia Amount (per hr)

- Scant: < 1 inch or 2.5 cm
- Small: < 4 inch or 10 cm
- Moderate: < 6 inch 15 cm
- Heavy: fully saturated pad

Breast Engorgement

- Breasts feels tender, warm, and taut
- Treatment: have baby suck frequently, breast massage, do interventions that increase milk flow, warm water, supportive bra
- Cabbage leaves can also treat breast engorgement

Meconium

- Stained fluid that is yellow/green or gold may indicate fetal stress

Maternity

Fetal Alcohol Syndrome

- Thin upper lip, flat nasal bridge, small for gestational age

Rhogam Factor

- Given IM at 28 weeks and 72 hours postpartum
- Only given mothers who are Rh-NEGATIVE (negative Coomb's test) who have a Rh-positive baby
 - If indirect Coomb's test is positive, that means the mother already has the antibody

Cephalhematoma (Caput Succinidanium)

- Edema that crosses the sutures lines
- Resolves on its own a few days after birth

Severe Pre-Eclampsia s/sx

HELLP

- Hemolysis
- Elevated Liver Function tests
- Low Platelet count

If → Then

- If a pregnant woman is choking and loses consciousness → then place in supine position with a wedge under right hip
 - Displaces uterus to the right and prevents supine hypotension
- If a woman comes to the hospital in active labor, the nurses first action is to listen to fetal heart rate
- If cord is prolapsed, cover it with sterile saline gauze to minimize risk for infection and drying of the cord
- If there is any dangerous fetal heart rate pattern, give mom oxygen
- If the baby is vertex (normal), the sounds are above the symphysis pubis
- If the baby is in a posterior presentation, heart sounds are heard at the sides
- If the baby is anterior, heart sounds are heard around midline (under umbilicus)
- If the baby is breech, heart sounds are high in the fundus (above umbilicus)

Labor Notes

- If pt has hypotension, bradypnea, or bradycardia: EMERGENCY
- Never check the monitor or machine as first action/priority. Always assess the pt first
- Determine cervical dilation before telling them to push. Pushing should begin at 10 cm (if started too early, cervix can become swollen → inhibiting full dilation)
- FHR is best heard through the fetal back in vertex

Newborn Notes

- Best way to warm a newborn → skin to skin contact

Additional Notes:

- Assess women in private if they are experiencing emotional or physical abuse from their partner

Mental Health

Pt SAFETY is #1 PRIORITY

Delirium

- Acute confusion or disorientation
- NI: Reassure pt, stay calm, provide safe environment

Dementia

- Chronic
- Redirect and orient pt to reality, show familiar objects, set daily schedule

Alzheimer's Disease

5 A's

- Amnesia (loss of memories)
- Aphasia (impacts ability to communicate)
- Apraxia (difficulty with skilled movement)
- Anomia (unable to recall known words)
- Agnosia (unable to identify objects, sounds, people)

Magical Thinking

- Pt believes their thoughts can control other people and events

Schizophrenia

- Impaired ability to perceive reality
- S/sx: hallucinations, delusions, disorganized speech and behavior, catatonia, negative symptoms
- NI: establish trust, provide safe environment, assist with ADL's and hygiene, sit with mute pt's, be nonjudgmental, use clear and simple sentences, avoid arguing with pt, praise socially acceptable behaviors
- Treatment: antipsychotics, individual and family therapy, social skills training

Positive Symptoms:

- Hallucinations
- Delusions

Negative Symptoms:

- Flat affect
- Anhedonia (lack of pleasure)
- Poverty of speech
- Movement disorders

Obsessive-Compulsive Disorders

- NI: actively listen to pt, acknowledge how the pt feels, provide empathy, avoid judgement
- Treatment: Antianxiety, SSRI's, tricyclic antidepressants

Post Traumatic Stress Disorder (PTSD)

- Triggered by traumatic event that can cause anxiety, flashbacks, and nightmares
- NI: actively listen to pt, assess suicide risk, help pt see the event objectively and identify areas of no control
- Treatment: antianxiety and antipsychotic meds

Personality Disorders

- Pt displays inappropriate emotional responses to stress, which is shown through anxious and fearful behaviors
- Includes: antisocial, narcissistic, borderline, schizotypal, avoidant, obsessive-compulsive
- NI: establish trust, protect pt from injury to self and others, help pt realize manipulative behaviors and set limits, focus on pt's strengths, encourage socialization

Munchausen Syndrome

- A person who pretends to be ill or inflicts self harm or illness in order to receive medical care or hospitalization

Mental Health

Munchausen by Proxy

- A person (usually a mother) intentionally causes or makes up an illness in another person (usually a child)

Tardive Dyskinesia

- Irreversible, involuntary movements of the tongue, face and extremities
- Due to prolonged use of antipsychotic meds

Depression

- Always assess pt, especially if they have a sudden increase in energy or mood (might be planning suicide)
- Symptoms can be manifested in somatic ways (pain, sleep issues, changes in appetite, low energy)
- NI: Directly ask pt how they feel and if they are suicidal, implement suicide precautions, monitor sleep and nutrition, encourage participation in activities, promote independence, sit quietly with pt if they do not want to talk, spend time with pt and keep your word (if you tell pt you will be back at 5pm, see them at 5pm)
- Suspect immediate suicide attempt if a depressed pt suddenly feels better → happy affect may imply pt is relieved to have a suicide plan and plans to follow through
- Signs pt is improving: pt takes interest in appearance, performs self-care activities,

Bipolar Disorder

- Bipolar I - mania with depressive episodes
- Bipolar II - milder episodes of hypomania that alternate with episodes of severe depression
- Manic s/sx: increased energy, impulsiveness, talkative, reduced need for sleep, excessive spending, delusions of grandeur or persecution
- NI: Ensure pt has adequate nutrition and rest, provide safe environment, decrease environmental stimulation, do not give attention to bizarre behavior, provide small frequent snacks/food (finger eating food)
- Treatment: Lithium, antipsychotics, sedatives, family therapy

Phobic Disorders

- Intense periods of fear and discomfort that can be incapacitating
- NI: Acknowledge fear, use systematic desensitization (exposure therapy) only AFTER trust is established

Anorexia Nervosa

- 2 Types: restrictive and binge/purging
- S/sx: weight loss (at least 15% of original body weight), distorted body image, excessive exercise, low self esteem, decreased BP and irregular HR, hair loss, dry skin, dehydration, no menstruation
- Absence of menstruation can lead to osteoporosis
- NI: monitor weight, vital signs, and electrolyte levels, provide supportive environment and structured mealtimes, set eating time limits, close assessment to food and fluid intake, watch for discarding of food, use positive reinforcement to build self esteem
- Treatment: Antidepressants, family therapy

Bulimia Nervosa

- S/sx: dental damage (d/t vomiting), sore throat, concerned with body shape (bulimics are not typically underweight), diarrhea, constipation, bloating
- NI: monitor weight, vital signs, and electrolyte levels, provide supportive environment and structured mealtimes, monitor for induced vomiting after meals, discuss strategies to stop laxative use and vomiting, provide positive reinforcement for increase self esteem
- Treatment: Antidepressants, family therapy

Alcoholism

5 D's

- Denial, destructive, demanding, domineering, dependency

Mental Health

Extrapyramidal Side Effects

- Rigidity, shuffling gait, tremors
- Administer an anticholinergic (ex: benztropine mesylate) to reverse side effects

Neuroleptic Malignant Syndrome (NMS)

- EMERGENCY caused by a reaction to antipsychotic meds
- NMS is like S&M: fever, diaphoresis, muscle rigidity, hypertension, tachycardia, tachypnea, drooling

Additional Notes:

- Do not argue with a pt about delusions. State things as a matter of fact and divert delusion to reality
- Do not tell pt you also hear voices
- Be supportive and nonjudgmental to all patients
- Adolescents are at an increased risk for suicide

Leadership

RN	LVN/PN	CNA/UAP
Responsible for assessment and teaching	Given patients with stable, expected outcomes	Given standard procedures
Reinforcing expected symptoms	Titrate oxygen based on unit protocols	Measure urine output
Calling family members about the patients status	Give oxygen via nasal cannula	Obtaining a non-sterile specimen Cannot obtain specimen from a foley catheter because it is a sterile procedure
Administer IV medications	Ostomy care	Colostomy bag change for a pt with an established stoma
Only RN's can handle blood and verify type	Give meds (no IV meds)	Remove foley cathetes
Must stay with pt during first 15 min of blood transfusion	Reinforcing teaching that have already been taught by the RN	Check blood glucose levels
	Administer tube feeds	ADL's
	NG tube insertion	Change linens
	Suctioning	Help w/ pt hygiene needs (shower, toilet)

RN

- Responsible for assessment, pt teaching, and nursing judgment
 - CANNOT delegate anything stated above

LVN/PN

- Similar to role of RN but CANNOT push or give any IV medications
- Can be given nursing interventions, but CANNOT assess pt

Advanced Directive

- Legal document that states a person's wishes regarding medical care
- Used in the event they are no longer able to make medical decisions for themselves due to serious injury/illness
- 2 types: living will & power of attorney
 - Living will: describes medical treatments a pt would/would not want to use used to keep them alive, as well as, organ donation and pain management
 - Power of attorney: Delegated person to make decisions for the pt when they are unable to do so. Also called health care agent/proxy. The person can be a spouse, family member, friend, or member of a faith community that is 18 years of age or older

Do not delegate what you can EAT

E Evaluate
A Assess
T Teach

Ethics

Fidelity

- Fulfilling commitments made to others
- Loyalty

Negligence

- Failure to adequately uphold the standard of care

Nonmaleficence

- Do no harm

Beneficence

- To do good

Justice

- Treating all patients fairly

Veracity

- Telling the truth

Pharmacology

Anticoagulant

-arin

Blood thinner

Ex: Heparin, Warfarin

Bronchodilator

-terol / -phylline

Dilates airways

Ex: Albuterol, Theophylline

Antiemetic

-azine

Treats nausea

Ex: Ondansetron,
Dexamethasone

Thrombolytic

-ase

Dissolves clots

Ex: Alteplase (TPA)
Streptokinase

Protein Pump Inhibitor

-prazole

Reduces gastric acid

Ex: Omeprazole,
Esomeprazole

H2 Blockers

-tidine

Reduces gastric acid

Ex: Famotidine,
Ranitidine

Antihistamine

-ine

Allergy relief

Ex: Diphenhydramine,
Loratadine

Antiviral

-vir

Treats viral infections

Ex: Acyclovir

Tetracyclines

-cycline

Antibiotic

Ex: Tetracycline,
Doxycycline

Tricyclic

-triptyline

Antidepressant

Ex: Amitriptyline,
Nortriptyline

SSRI

-pram / -ine

**Antidepressant &
Anxiolytic**

Ex: Escitalopram,
Sertraline

Aminoglycoside

-mycin

Antibiotic

Ex: Vancomycin,
Erythromycin

Fluoroquinolones

-floxacin

Antibiotic

Ex: Levofloxacin

Pharmacology

Antihypertensives - decrease BP

Cardiac (ABCDD)

- A** Ace Inhibitor *decreases BP
 - Ends in -pril
 - Ex: Lisinopril
- B** Beta Blocker *decreases HR
 - Ends in -lol
 - Ex: Atenolol
- C** Calcium Channel Blocker *decreases BP & HR
 - Ends in -dipine, -zem, amil
 - Ex: Nifedipine, Diltiazem, Verapamil
- D** Digoxin *increase contractility
 - Cardiac glycoside → deep contraction and increases contractility
 - Hold if apical pulse is <60, Potassium under 3.5, Digoxin toxicity >2 (signs and symptoms: vision changes, nausea/vomiting, fatigue, dizziness)
- D** Diuretics *decreases BP
 - Loop or thiazide
 - Furosemide causes dehydration (furosemide=body is dried)
 - Potassium WASTING → teach pt. to increase intake of potassium foods

Cardiac Notes:

- Vasopressin - think press in/vasoconstrict
- If on oral anticoagulants, glucagon (controls glucose levels) may increase effect → risk for bleeding

Morphine

- Preload reducer = less blood in right atrium

Nitroglycerine

- Vasodilator → increases oxygen, reduces preload

Analgesics

- Acetaminophen (Tylenol) - used for fever and pain relief

NSAIDS

- Used for inflammation, fever, and pain relief
- Types
 - Indomethacin
 - Ibuprofen
 - Naproxen (treats joint pain and inflammation)
- Side effects
 - GI toxicity: black tarry stools, dyspepsia and pain if taken on empty stomach
 - Kidney injury
 - HTN and HF: cause fluid retention
 - Bleeding risk
- Monitor liver enzymes

Antivirals

- Help body fight off virus; not a cure
- Acyclovir (Zovirax)

Antibiotics

- Destroys or slows down growth of bacteria
- NI: always obtain cultures before starting antibiotics

Pharmacology

Anticoagulant

- Slows down the body's process of making clots
- Heparin, Warfarin, Lovenox, Plavix

Antiplatelet

- Prevent platelets from clumping together to form a clot
- Aspirin
- Given to pts who have had a MI or stroke

Bronchodilators

- Relaxes bronchi → allows air to come in and out of the lungs & assists with excreting mucus
- Used in inhalers or nebulizers
- Ipratropium (Atrovent), budesonide (Pulmicort)

Antacids

- Reduce gastric acidity by raising pH of the stomach contents and protecting stomach mucosa
- Treats heartburn, ulcers, gastritis, esophagitis, hyperphosphatemia

Antiemetics

- Reduces nausea, vomiting, and motion sickness
- Ondansetron (Zofran)

Antihistamines

- Reduces s/sx of allergies; some cause drowsiness
- Cetirizine (Zyrtec), fexofenadine (Allegra)

Steroid Side Effects

6 S's

- Sugar - hyperglycemia
- Salt - salt and water retention → HTN
- Sex - decreased libido
- Sad - depression
- Soggy bones - osteoporosis
- Sick - decreased immunity

Antidepressants

SSRI's

- Antidepressant that is effective for sadness, compulsions, and panic

MAOI's - antidepressant

- Think PANAMA and arrrr (Pirates say arrr → Pirates take MAOI's when they are depressed)
 - PA- parnate, NA - nardil, MA - marplan

Antipsychotics

- Treats psychosis and decreases agitation and confusion
- Typical - positive symptoms
- Atypical - negative symptoms
- Can take several weeks to reach desired effect

When to Take

- Diuretics → AM
- Steroids → AM
- Growth Hormone → PM

Additional Notes

- Pt's taking digoxin or potassium supplements: avoid salt substitutes (contain potassium)
- Atropine is also used for pupil dilation and paralysis in preparation for examination or surgery
- Do NOT give morphine sulfate to patients with pancreatitis. Give pt demerol
- Pt's taking digoxin and/or lasix are at risk for hypokalemia. Ensure pt is receiving enough potassium.
 - Hypokalemia → Digoxin toxicity → hyperkalemia

Emergency Medications

- L** Lidocaine
- E** Epinephrine
- A** Atropine Sulfate
- N** Narcan

Good luck on the NCLEX!

Notes

References

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